Causal Analysis Report

bpic18 Parcel document.xes.gz-dfg

Decision points:	131505
Choice sets:	6
Transitions fitted:	21
Significant (p<0.05):	1
Generated:	2026-06-10 09:22

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

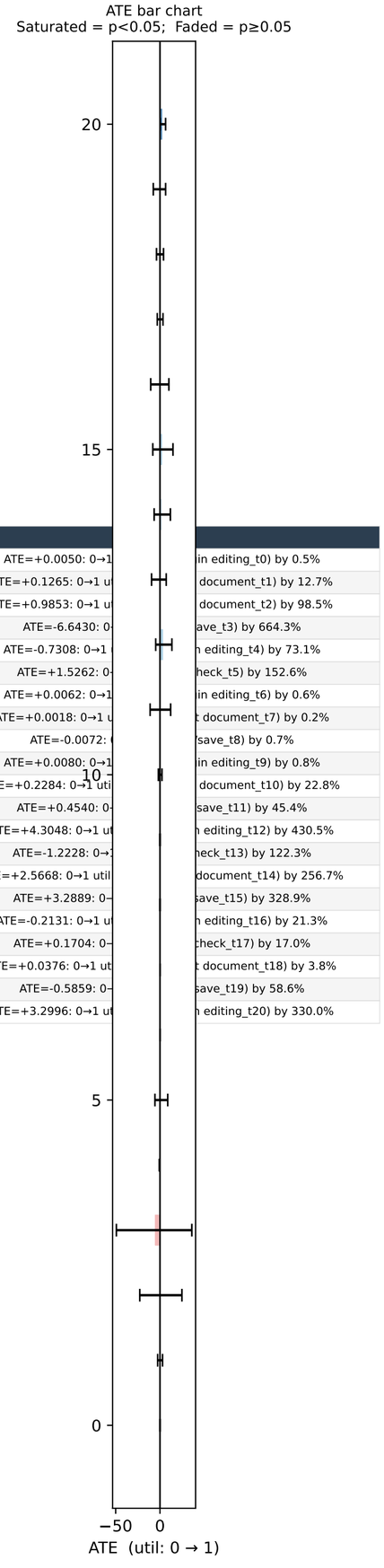
choice_set	choice_set_labels	n	choices_seen	coef	ci_95	perm_p_value	reject_H0	note
(0, 1)	('begin editing_t0', 'insert document_t1')	14750	0, 1	-5.4825	[-5.901, -5.079]	0.0	True	higher util → t0 (begin editing_t0) over t1 (insert document_t1)
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	1.6807	[1.212, 2.164]	0.0	True	higher util → t5 (check_t5) over t2 (insert document_t2)
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	-2.1665	[-2.449, -1.124]	0.0	True	higher util → t6 (begin editing_t6) over t8 (save_t8)
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	-5.8165	[-6.256, -5.399]	0.0	True	higher util → t9 (begin editing_t9) over t13 (check_t13)
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	0.3087	[-0.347, 0.935]	0.0175	True	higher util → t17 (check_t17) over t14 (insert document_t14)
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	18.1106	[16.729, 19.366]	0.0	True	higher util → t20 (finish editing_t20) over t18 (insert document_t18)

Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

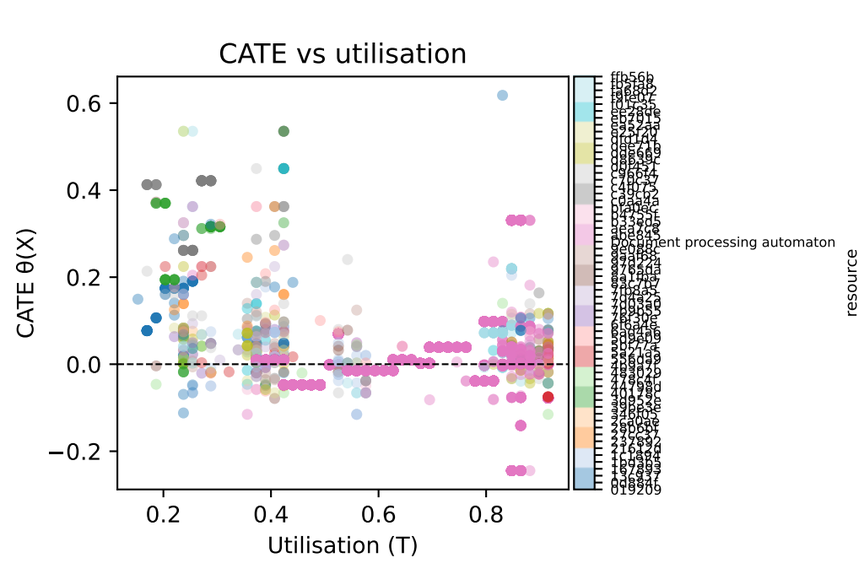
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1)	('begin editing_t0', 'insert document_t1')	14750	0, 1	0	begin editing_t0	insert document_t1	0.027188	0.026884	[-0.0255, 0.0799]	0.3119	False	θ=+0.0272: 1-unit ↑ util ↑ P(t0/begin editing_t0) by 2.7%
(0, 1)	('begin editing_t0', 'insert document_t1')	14750	0, 1	1	insert document_t1	begin editing_t0	-0.043888	0.029684	[-0.1021, 0.0143]	0.1393	False	θ=-0.0439: 1-unit ↑ util ↓ P(t1/insert document_t1) by 4.4%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	2	insert document_t2	check_t5, finish editing_t4, save_t3	-4.440056	0.291344	[-5.0111, -3.8690]	0.0	True	θ=-4.4401: 1-unit ↑ util ↓ P(t2/insert document_t2) by 444.0%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	3	save_t3	check_t5, finish editing_t4, insert document_t2	1.090755	0.170718	[0.7561, 1.4254]	0.0	True	θ=+1.0908: 1-unit ↑ util ↑ P(t3/save_t3) by 109.1%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	4	finish editing_t4	check_t5, insert document_t2, save_t3	-0.253406	0.08095	[-0.4121, -0.0947]	0.0017	True	θ=-0.2534: 1-unit ↑ util ↓ P(t4/finish editing_t4) by 25.3%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	5	check_t5	finish editing_t4, insert document_t2, save_t3	2.412516	0.155058	[2.1086, 2.7164]	0.0	True	θ=+2.4125: 1-unit ↑ util ↑ P(t5/check_t5) by 241.3%
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	6	begin editing_t6	insert document_t7, save_t8	-0.017686	0.006741	[-0.0309, -0.0045]	0.0087	True	θ=-0.0177: 1-unit ↑ util ↓ P(t6/begin editing_t6) by 1.8%
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	7	insert document_t7	begin editing_t6, save_t8	0.000361	0.001617	[-0.0028, 0.0035]	0.8231	False	θ=+0.0004: 1-unit ↑ util ↑ P(t7/insert document_t7) by 0.0%
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	8	save_t8	begin editing_t6, insert document_t7	0.010771	0.016884	[-0.0223, 0.0439]	0.5235	False	θ=+0.0108: 1-unit ↑ util ↑ P(t8/save_t8) by 1.1%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	9	begin editing_t9	check_t13, finish editing_t12, insert document_t10, save_t11	0.034869	0.0115	[0.0123, 0.0574]	0.0024	True	θ=+0.0349: 1-unit ↑ util ↑ P(t9/begin editing_t9) by 3.5%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	10	insert document_t10	begin editing_t9, check_t13, finish editing_t12, save_t11	0.486249	0.035594	[0.4165, 0.5560]	0.0	True	θ=+0.4862: 1-unit ↑ util ↑ P(t10/insert document_t10) by 48.6%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	11	save_t11	begin editing_t9, check_t13, finish editing_t12, insert document_t10	-0.831591	0.302714	[-1.4249, -0.2383]	0.006	True	θ=-0.8316: 1-unit ↑ util ↓ P(t11/save_t11) by 83.2%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	12	finish editing_t12	begin editing_t9, check_t13, insert document_t10, save_t11	3.680443	0.185216	[3.3174, 4.0435]	0.0	True	θ=+3.6804: 1-unit ↑ util ↑ P(t12/finish editing_t12) by 368.0%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	13	check_t13	begin editing_t9, finish editing_t12, insert document_t10, save_t11	0.399404	0.15772	[0.0903, 0.7085]	0.0113	True	θ=+0.3994: 1-unit ↑ util ↑ P(t13/check_t13) by 39.9%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	14	insert document_t14	check_t17, finish editing_t16, save_t15	3.455677	0.22129	[3.0219, 3.8894]	0.0	True	θ=+3.4557: 1-unit ↑ util ↑ P(t14/insert document_t14) by 345.6%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	15	save_t15	check_t17, finish editing_t16, insert document_t14	2.217177	0.209054	[1.8074, 2.6269]	0.0	True	θ=+2.2172: 1-unit ↑ util ↑ P(t15/save_t15) by 221.7%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	16	finish editing_t16	check_t17, insert document_t14, save_t15	0.016879	0.19073	[-0.3570, 0.3907]	0.9295	False	θ=+0.0169: 1-unit ↑ util ↑ P(t16/finish editing_t16) by 1.7%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	17	check_t17	finish editing_t16, insert document_t14, save_t15	0.189124	0.052479	[0.0863, 0.2920]	0.0003	True	θ=+0.1891: 1-unit ↑ util ↑ P(t17/check_t17) by 18.9%
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	18	insert document_t18	finish editing_t20, save_t19	0.906931	0.161595	[0.5902, 1.2237]	0.0	True	θ=+0.9069: 1-unit ↑ util ↑ P(t18/insert document_t18) by 90.7%
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	19	save_t19	finish editing_t20, insert document_t18	0.084247	0.237091	[-0.3805, 0.5489]	0.7223	False	θ=+0.0842: 1-unit ↑ util ↑ P(t19/save_t19) by 8.4%
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	20	finish editing_t20	insert document_t18, save_t19	3.121897	0.123717	[2.8794, 3.3644]	0.0	True	θ=+3.1219: 1-unit ↑ util ↑ P(t20/finish editing_t20) by 312.2%

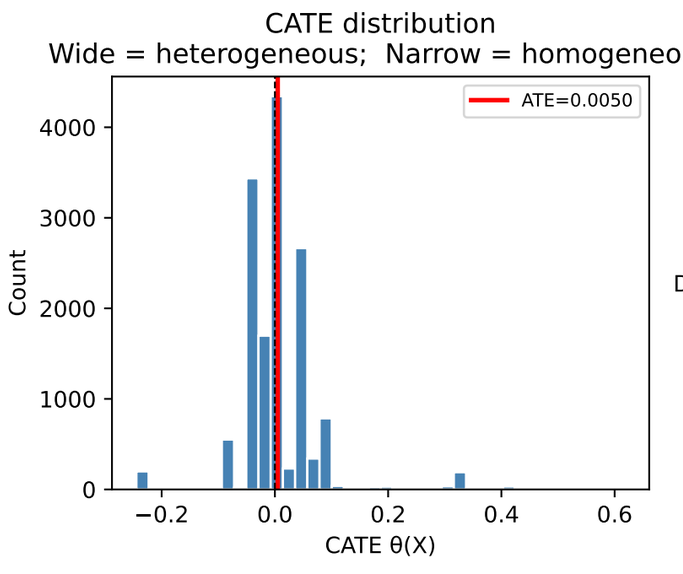
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

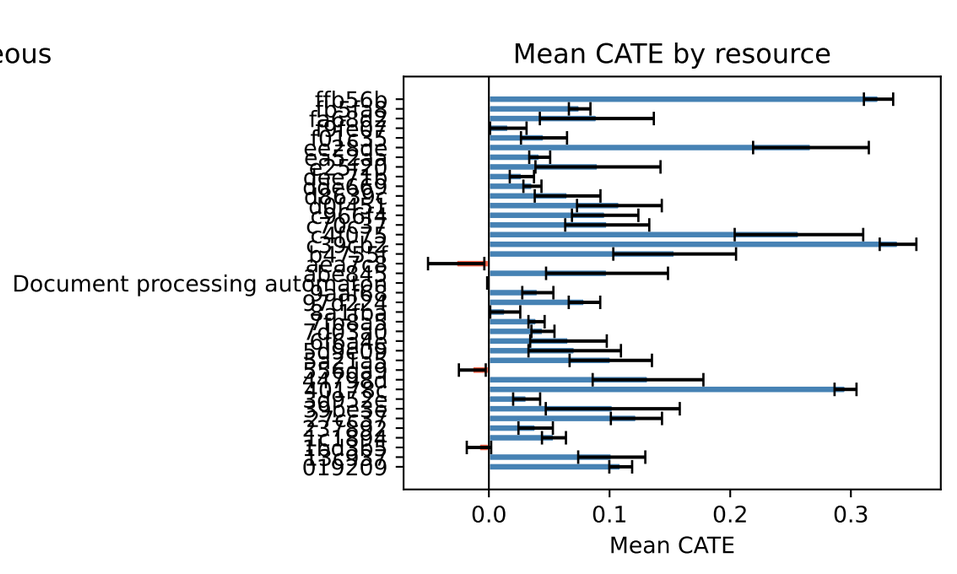
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	
(0, 1)	('begin editing_t0', 'insert document_t1')	14750	0, 1	0	begin editing_t0	insert document_t1	0.00497	[-0.2006, 0.2106]	0.9622	False	ATE=+0.0050: 0→1 in editing_t0) by 0.5%
(0, 1)	('begin editing_t0', 'insert document_t1')	14750	0, 1	1	insert document_t1	begin editing_t0	0.126508	[-2.6456, 2.8986]	0.9287	False	ATE=+0.1265: 0→1 u document_t1) by 12.7%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	2	insert document_t2	check_t5, finish editing_t4, save_t3	0.985313	[-22.7953, 24.7659]	0.9353	False	ATE=+0.9853: 0→1 u document_t2) by 98.5%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	3	save_t3	check_t5, finish editing_t4, insert document_t2	-6.643038	[-49.2084, 35.9224]	0.7597	False	ATE=-6.6430: 0 ave_t3) by 664.3%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	4	finish editing_t4	check_t5, insert document_t2, save_t3	-0.730771	[-0.7308, -0.7308]	1.0	False	ATE=-0.7308: 0→1 editing_t4) by 73.1%
(2, 3, 4, 5)	('insert document_t2', 'save_t3', 'finish editing_t4', 'check_t5')	14749	2, 3, 4, 5	5	check_t5	finish editing_t4, insert document_t2, save_t3	1.526212	[-5.6868, 8.7393]	0.6783	False	ATE=+1.5262: 0 heck_t5) by 152.6%
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	6	begin editing_t6	insert document_t7, save_t8	0.006157	[-0.0846, 0.0969]	0.8942	False	ATE=+0.0062: 0→1 in editing_t6) by 0.6%
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	7	insert document_t7	begin editing_t6, save_t8	0.001847	[-0.0485, 0.0522]	0.9427	False	ATE=+0.0018: 0→1 u t document_t7) by 0.2%
(6, 7, 8)	('begin editing_t6', 'insert document_t7', 'save_t8')	9601	6, 7, 8	8	save_t8	begin editing_t6, insert document_t7	-0.007172	[-0.2402, 0.2258]	0.9519	False	ATE=-0.0072: (save_t8) by 0.7%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	9	begin editing_t9	check_t13, finish editing_t12, insert document_t10, save_t11	0.007995	[-0.2868, 0.3028]	0.9576	False	ATE=+0.0080: 0→1 in editing_t9) by 0.8%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	10	insert document_t10	begin editing_t9, check_t13, finish editing_t12, save_t11	0.228443	[-1.8140, 2.2709]	0.8265	False	ATE=+0.2284: 0→1 uti document_t10) by 22.8%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	11	save_t11	begin editing_t9, check_t13, finish editing_t12, insert document_t10	0.454035	[-11.1042, 12.0123]	0.9386	False	ATE=+0.4540: 0 save_t11) by 45.4%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	12	finish editing_t12	begin editing_t9, check_t13, insert document_t10, save_t11	4.304847	[-4.8301, 13.4398]	0.3557	False	ATE=+4.3048: 0→1 ut editing_t12) by 430.5%
(9, 10, 11, 12, 13)	('begin editing_t9', 'insert document_t10', 'save_t11', 'finish editing_t12', 'check_t13')	34043	9, 10, 11, 12, 13	13	check_t13	begin editing_t9, finish editing_t12, insert document_t10, save_t11	-1.222846	[-9.5052, 7.0595]	0.7723	False	ATE=-1.2228: 0→ heck_t13) by 122.3%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	14	insert document_t14	check_t17, finish editing_t16, save_t15	2.566755	[-6.5863, 11.7198]	0.5826	False	ATE=+2.5668: 0→1 util document_t14) by 256.7%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	15	save_t15	check_t17, finish editing_t16, insert document_t14	3.288921	[-8.1008, 14.6787]	0.5714	False	ATE=+3.2889: 0→ save_t15) by 328.9%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	16	finish editing_t16	check_t17, insert document_t14, save_t15	-0.213058	[-10.4572, 10.0310]	0.9675	False	ATE=-0.2131: 0→1 u editing_t16) by 21.3%
(14, 15, 16, 17, 21)	('insert document_t14', 'save_t15', 'finish editing_t16', 'check_t17', 'τ_t21')	31436	14, 15, 16, 17	17	check_t17	finish editing_t16, insert document_t14, save_t15	0.170446	[-3.0318, 3.3727]	0.9169	False	ATE=+0.1704: 0→ check_t17) by 17.0%
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	18	insert document_t18	finish editing_t20, save_t19	0.037646	[-3.9205, 3.9958]	0.9851	False	ATE=+0.0376: 0→1 ut t document_t18) by 3.8%
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	19	save_t19	finish editing_t20, insert document_t18	-0.585879	[-7.5382, 6.3664]	0.8688	False	ATE=-0.5859: 0→ save_t19) by 58.6%
(18, 19, 20)	('insert document_t18', 'save_t19', 'finish editing_t20')	12135	18, 19, 20	20	finish editing_t20	insert document_t18, save_t19	3.29958	[0.2565, 6.3427]	0.0336	True	ATE=+3.2996: 0→1 ut editing_t20) by 330.0%

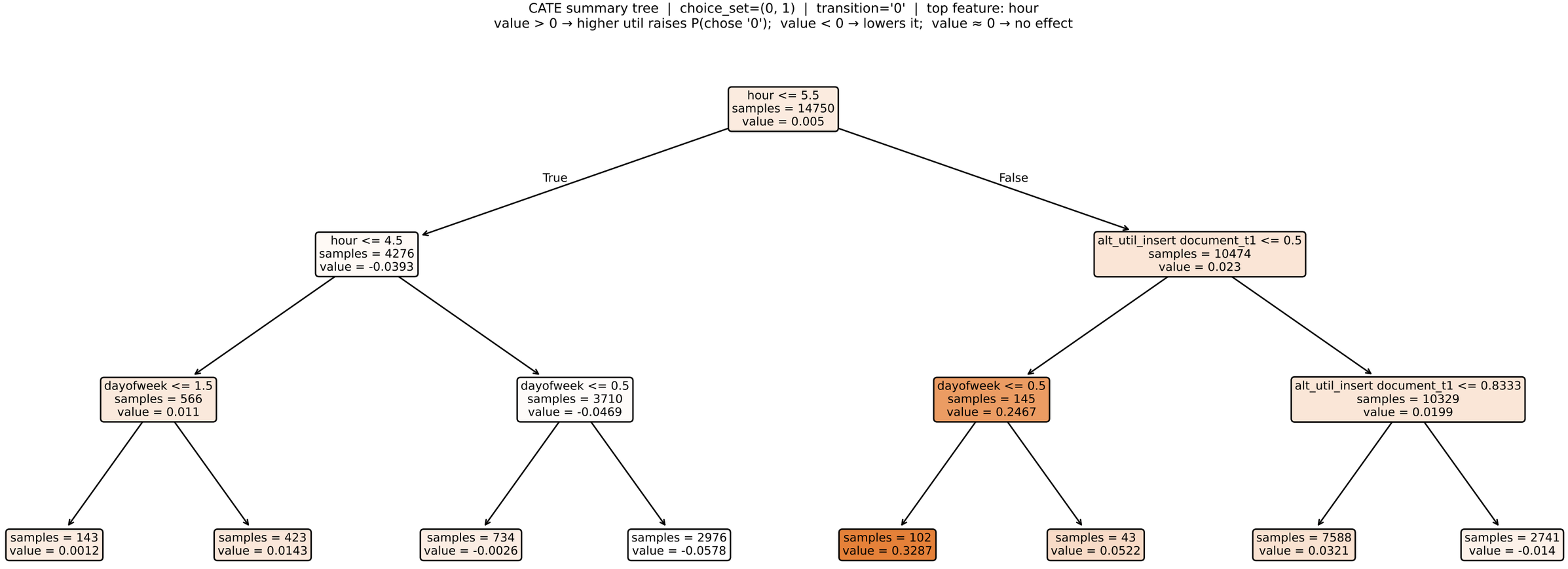


CATE — choice_set=(0, 1) | transition='0' | n=14750 | ATE=0.0050 std=0.0732

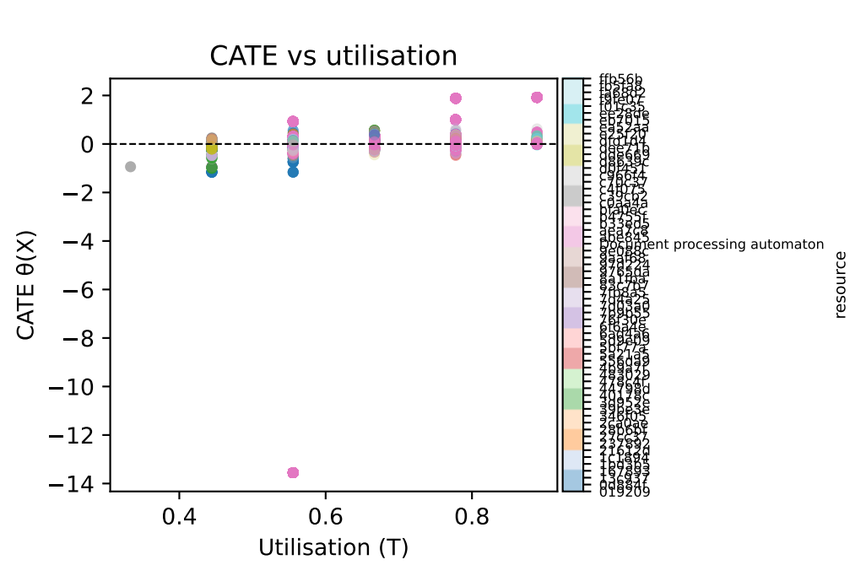


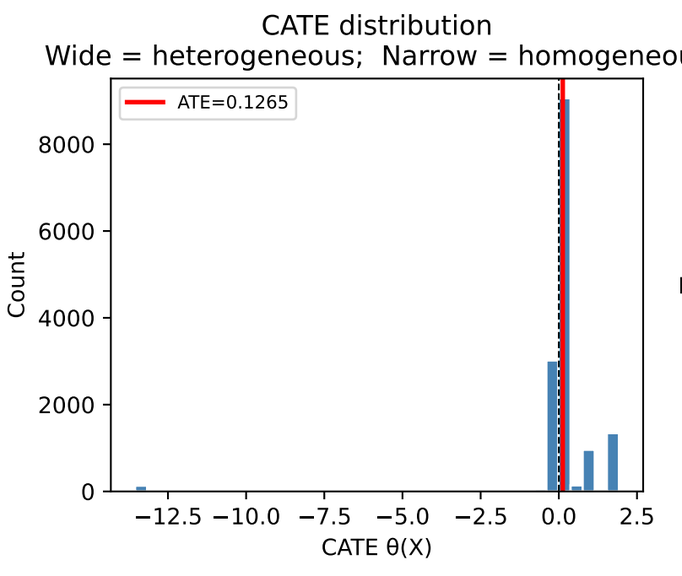


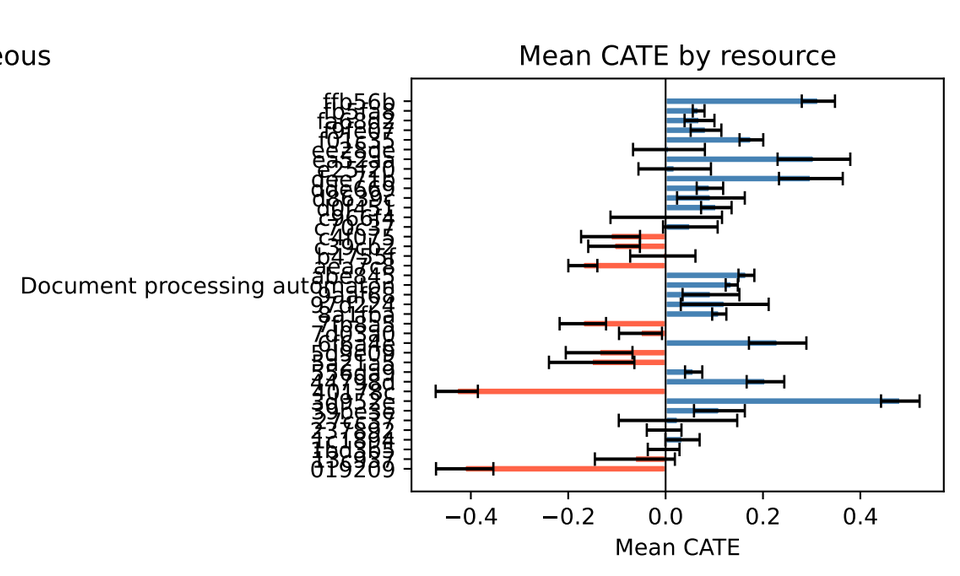


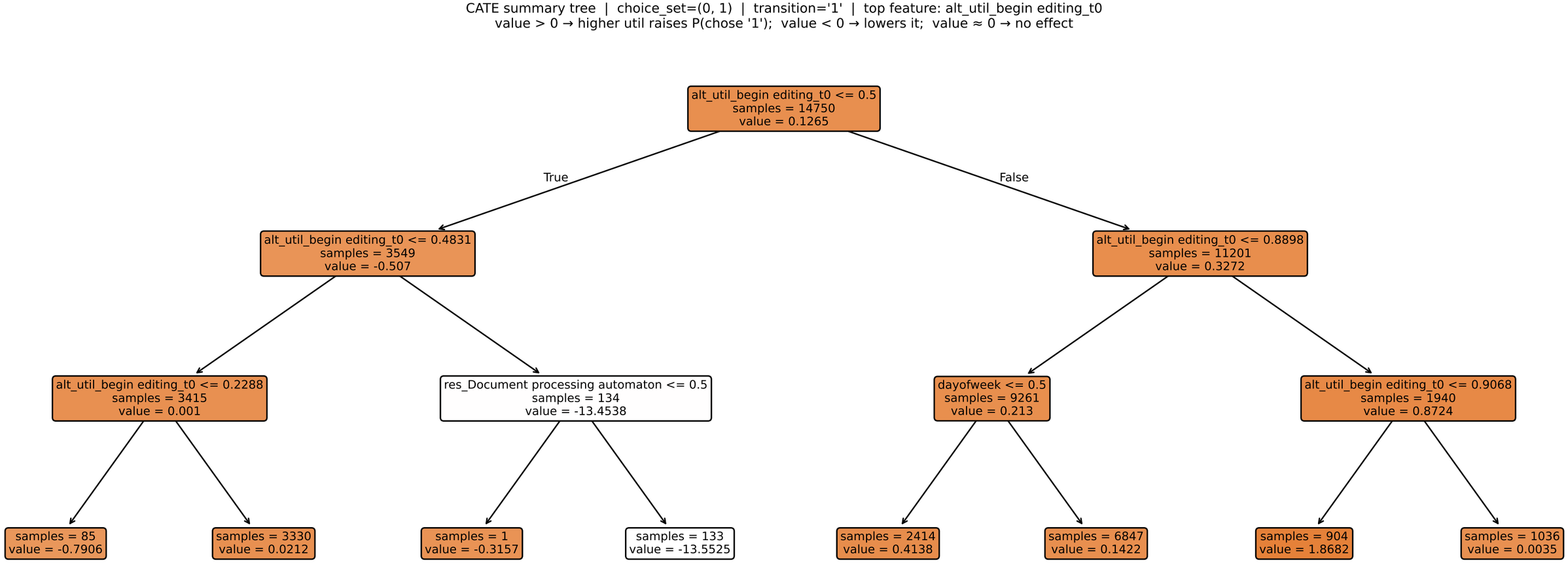

CATE summary tree | choice_set=(0, 1) | transition='0' | top feature: hour
value > 0 → higher util raises P(chose '0'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 5.5
samples = 14750
value = 0.005
True
False
hour <= 4.5
samples = 4276
value = -0.0393
alt_util_insert document_t1 <= 0.5
samples = 10474
value = 0.023
dayofweek <= 1.5
samples = 566
value = 0.011
dayofweek <= 0.5
samples = 3710
value = -0.0469
dayofweek <= 0.5
samples = 145
value = 0.2467
alt_util_insert document_t1 <= 0.8333
samples = 10329
value = 0.0199
samples = 143
value = 0.0012
samples = 423
value = 0.0143
samples = 734
value = -0.0026
samples = 2976
value = -0.0578
samples = 102
value = 0.3287
samples = 43
value = 0.0522
samples = 7588
value = 0.0321
samples = 2741
value = -0.014

CATE — choice_set=(0, 1) | transition='1' | n=14750 | ATE=0.1265 std=1.4331

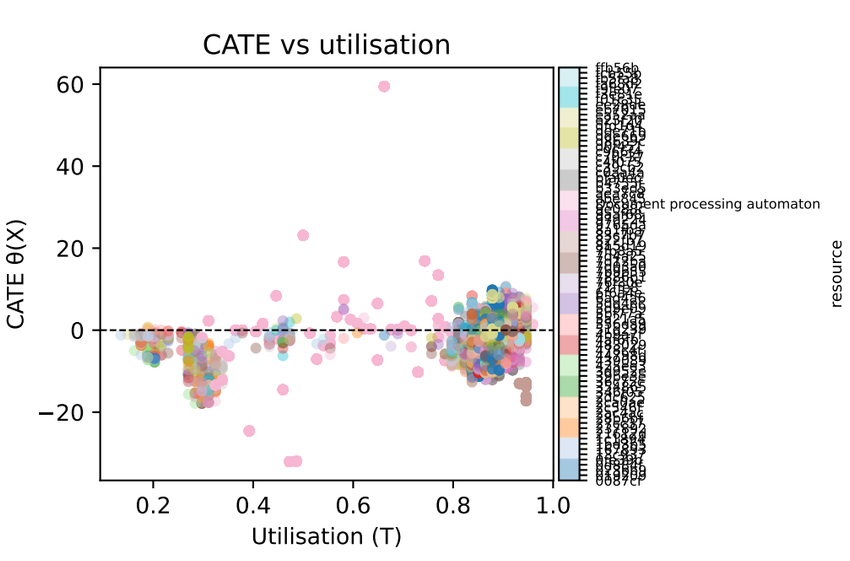


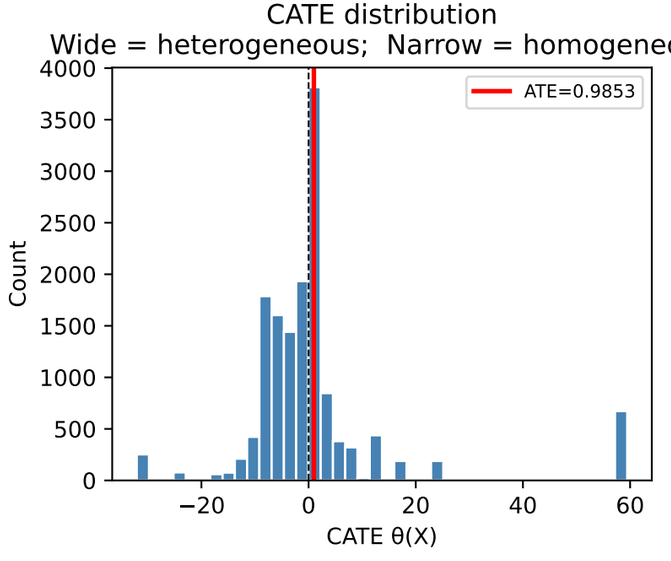


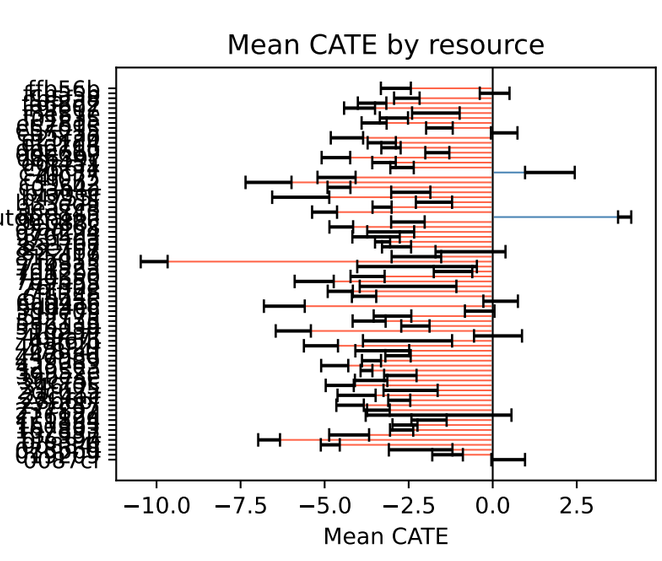


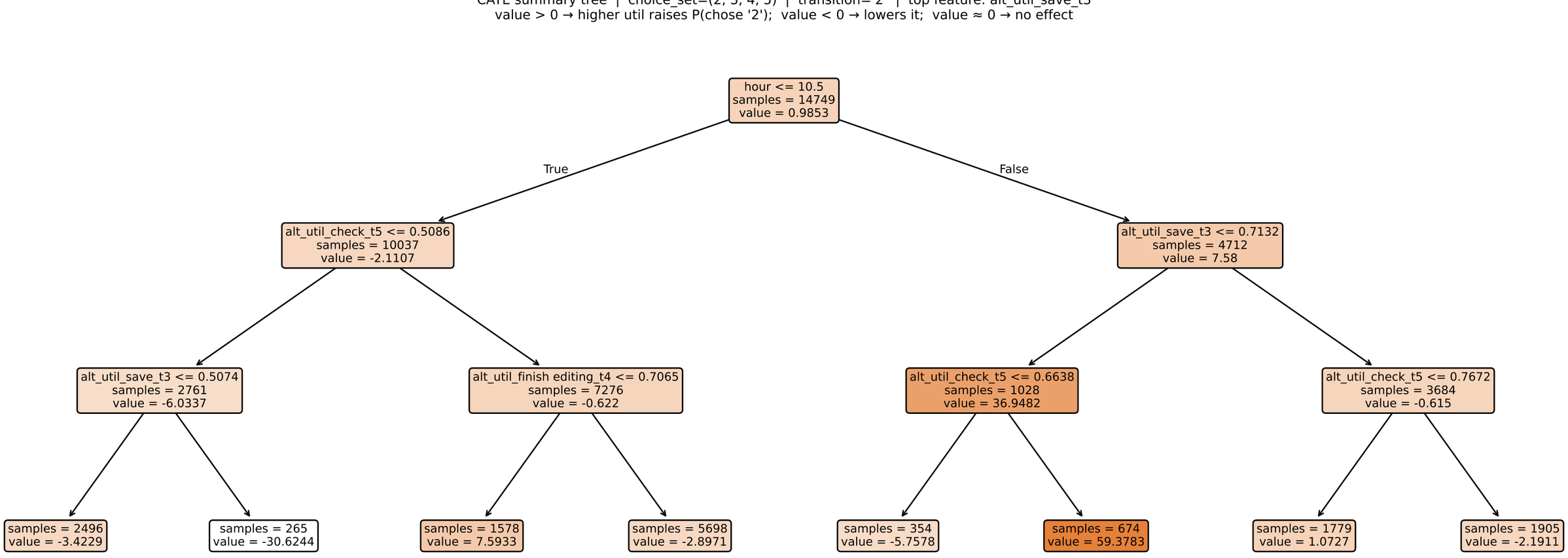

CATE summary tree | choice_set=(0, 1) | transition='1' | top feature: alt_util_begin editing_t0
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_begin editing_t0 <= 0.5
samples = 14750
value = 0.1265
True
False
alt_util_begin editing_t0 <= 0.4831
samples = 3549
value = -0.507
alt_util_begin editing_t0 <= 0.8898
samples = 11201
value = 0.3272
alt_util_begin editing_t0 <= 0.2288
samples = 3415
value = 0.001
res_Document processing automaton <= 0.5
samples = 134
value = -13.4538
dayofweek <= 0.5
samples = 9261
value = 0.213
alt_util_begin editing_t0 <= 0.9068
samples = 1940
value = 0.8724
samples = 85
value = -0.7906
samples = 3330
value = 0.0212
samples = 1
value = -0.3157
samples = 133
value = -13.5525
samples = 2414
value = 0.4138
samples = 6847
value = 0.1422
samples = 904
value = 1.8682
samples = 1036
value = 0.0035

CATE — choice_set=(2, 3, 4, 5) | transition='2' | n=14749 | ATE=0.9853 std=14.8282

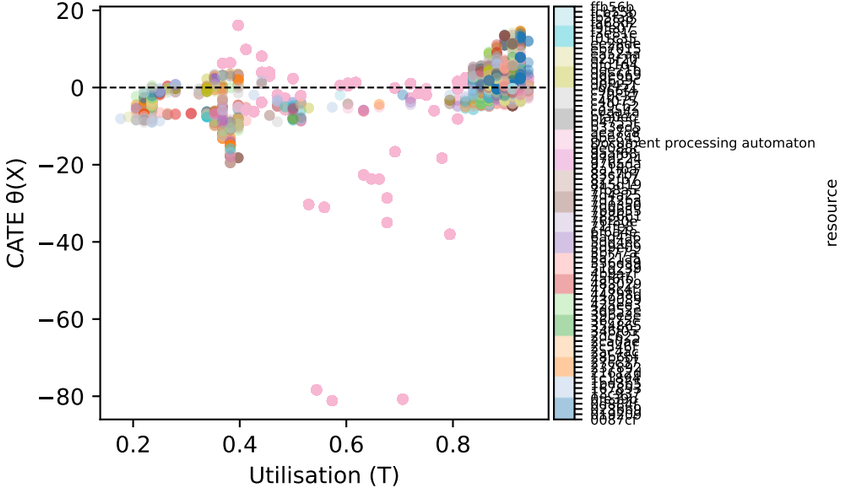


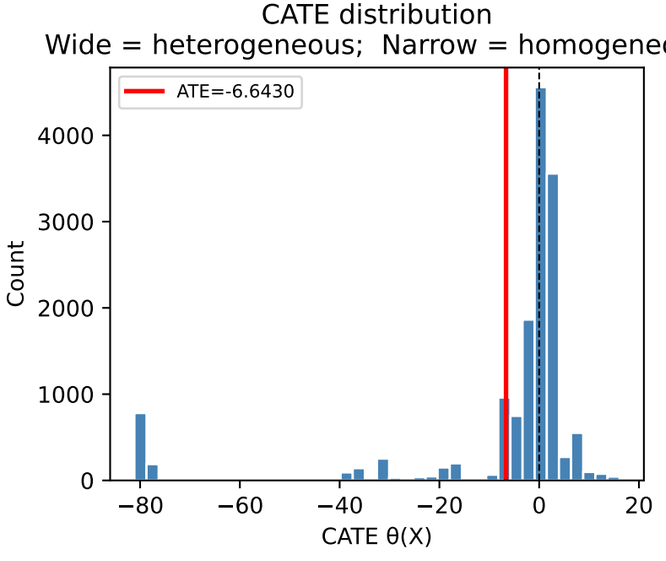


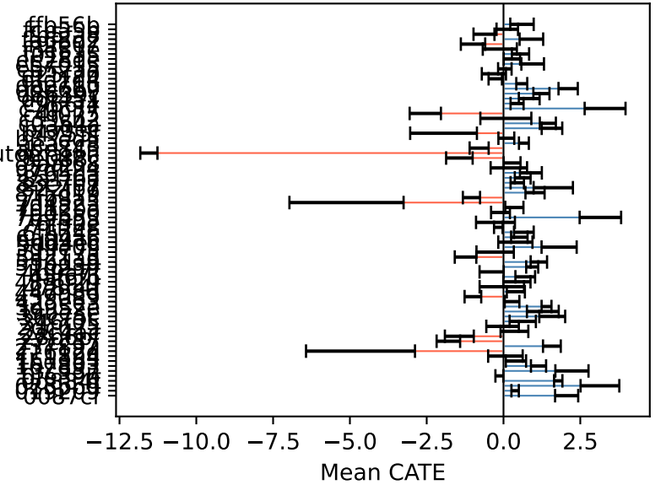


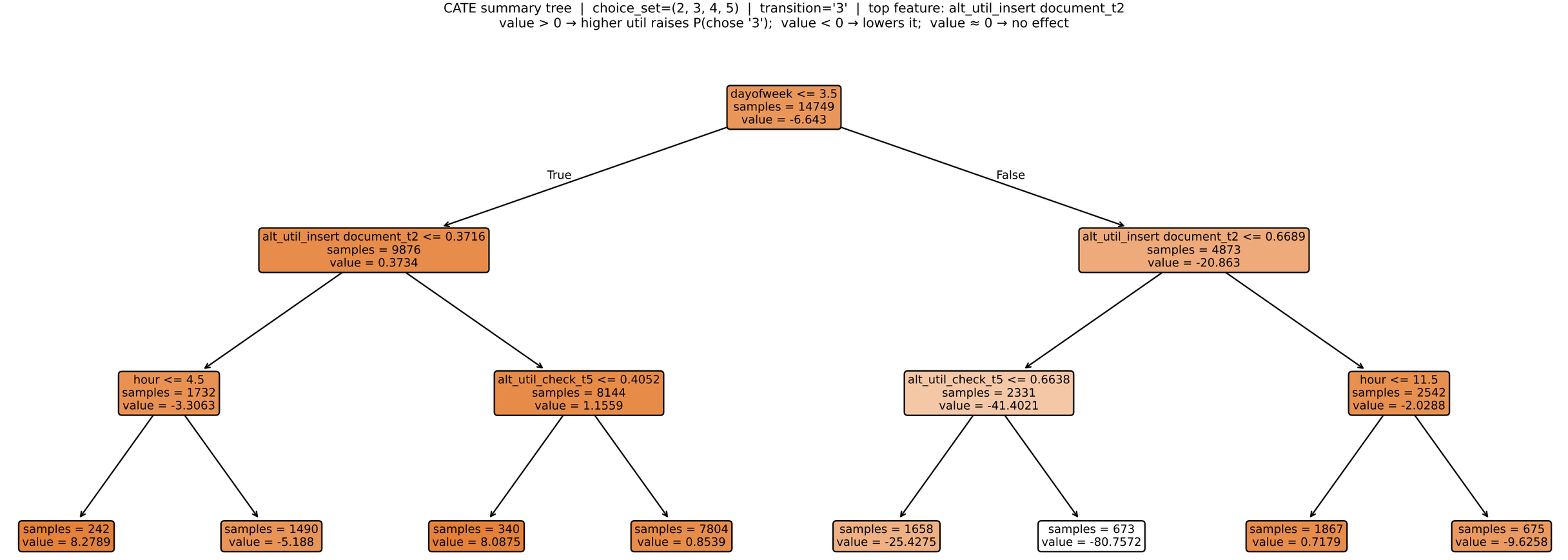

CATE summary tree | choice_set=(2, 3, 4, 5) | transition='2' | top feature: alt_util_save_t3
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 10.5
samples = 14749
value = 0.9853
True
False
alt_util_check_t5 <= 0.5086
samples = 10037
value = -2.1107
alt_util_save_t3 <= 0.7132
samples = 4712
value = 7.58
alt_util_save_t3 <= 0.5074
samples = 2761
value = -6.0337
alt_util_finish editing_t4 <= 0.7065
samples = 7276
value = -0.622
alt_util_check_t5 <= 0.6638
samples = 1028
value = 36.9482
alt_util_check_t5 <= 0.7672
samples = 3684
value = -0.615
samples = 2496
value = -3.4229
samples = 265
value = -30.6244
samples = 1578
value = 7.5933
samples = 5698
value = -2.8971
samples = 354
value = -5.7578
samples = 674
value = 59.3783
samples = 1779
value = 1.0727
samples = 1905
value = -2.1911

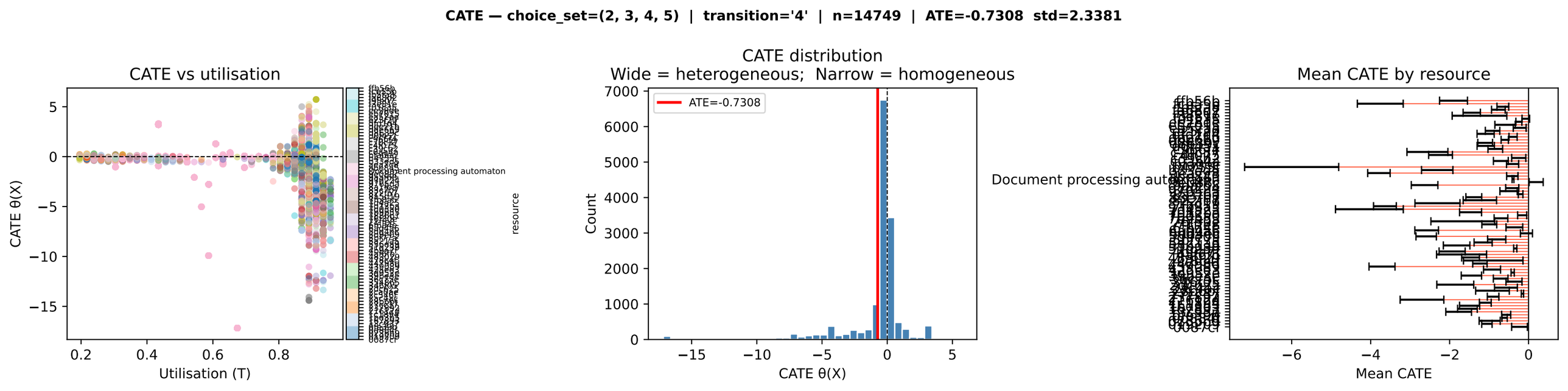
CATE — choice_set=(2, 3, 4, 5) | transition='3' | n=14749 | ATE=-6.6430 std=20.9937

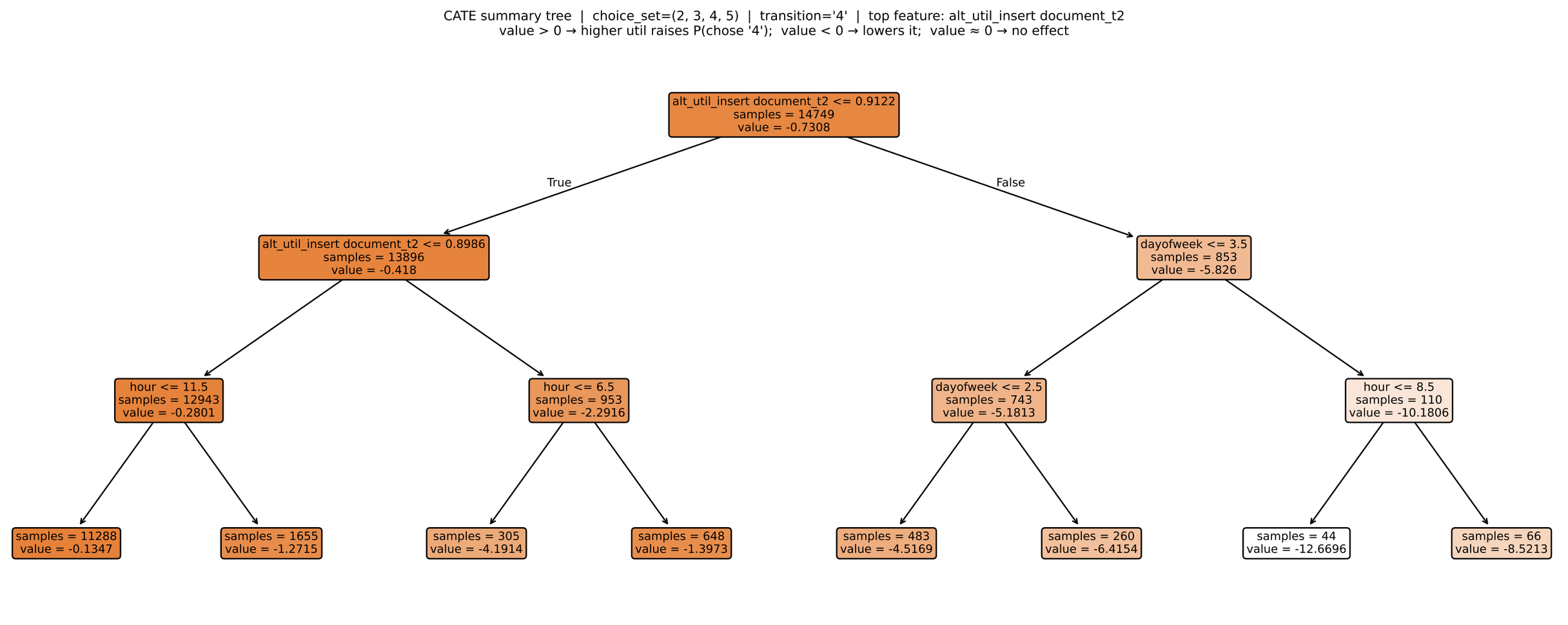




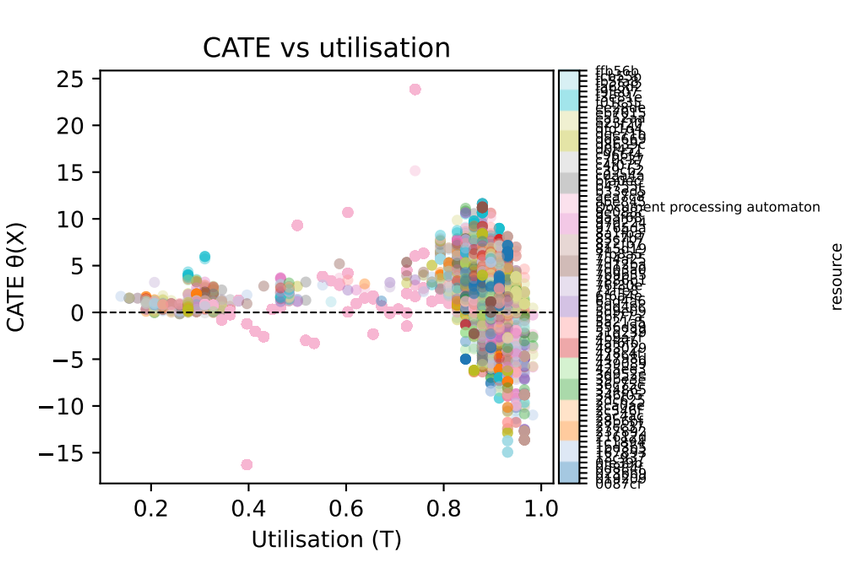


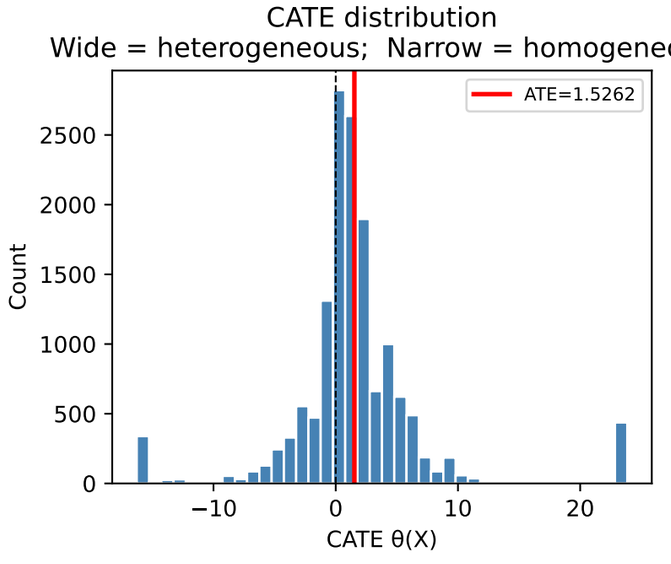

CATE summary tree | choice_set=(2, 3, 4, 5) | transition='3' | top feature: alt_util_insert document_t2
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 3.5
samples = 14749
value = -6.643
True
False
alt_util_insert document_t2 <= 0.3716
samples = 9876
value = 0.3734
alt_util_insert document_t2 <= 0.6689
samples = 4873
value = -20.863
hour <= 4.5
samples = 1732
value = -3.3063
alt_util_check_t5 <= 0.4052
samples = 8144
value = 1.1559
alt_util_check_t5 <= 0.6638
samples = 2331
value = -41.4021
hour <= 11.5
samples = 2542
value = -2.0288
samples = 242
value = 8.2789
samples = 1490
value = -5.188
samples = 340
value = 8.0875
samples = 7804
value = 0.8539
samples = 1658
value = -25.4275
samples = 673
value = -80.7572
samples = 1867
value = 0.7179
samples = 675
value = -9.6258

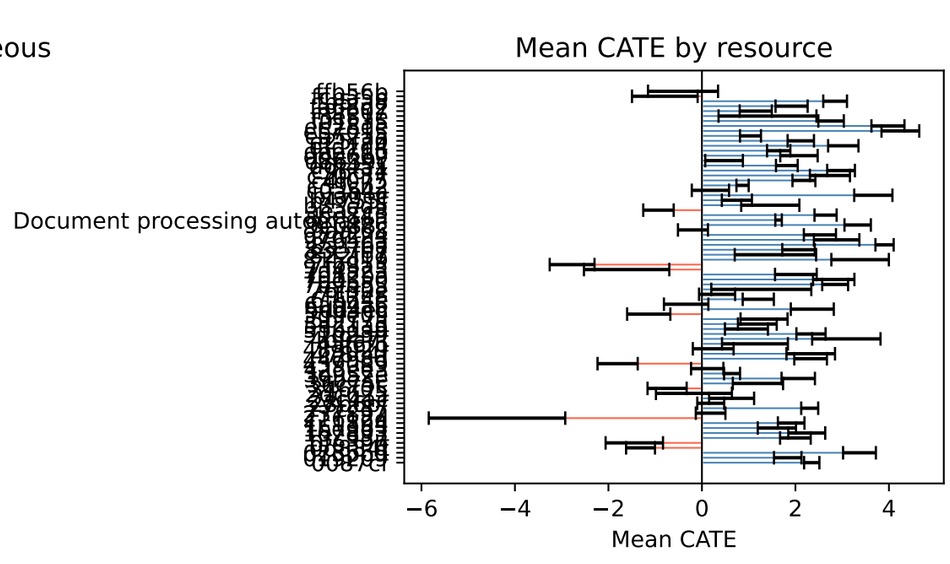

CATE — choice_set=(2, 3, 4, 5) | transition='4' | n=14749 | ATE=-0.7308 std=2.3381
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
Document processing automaton
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.7308
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

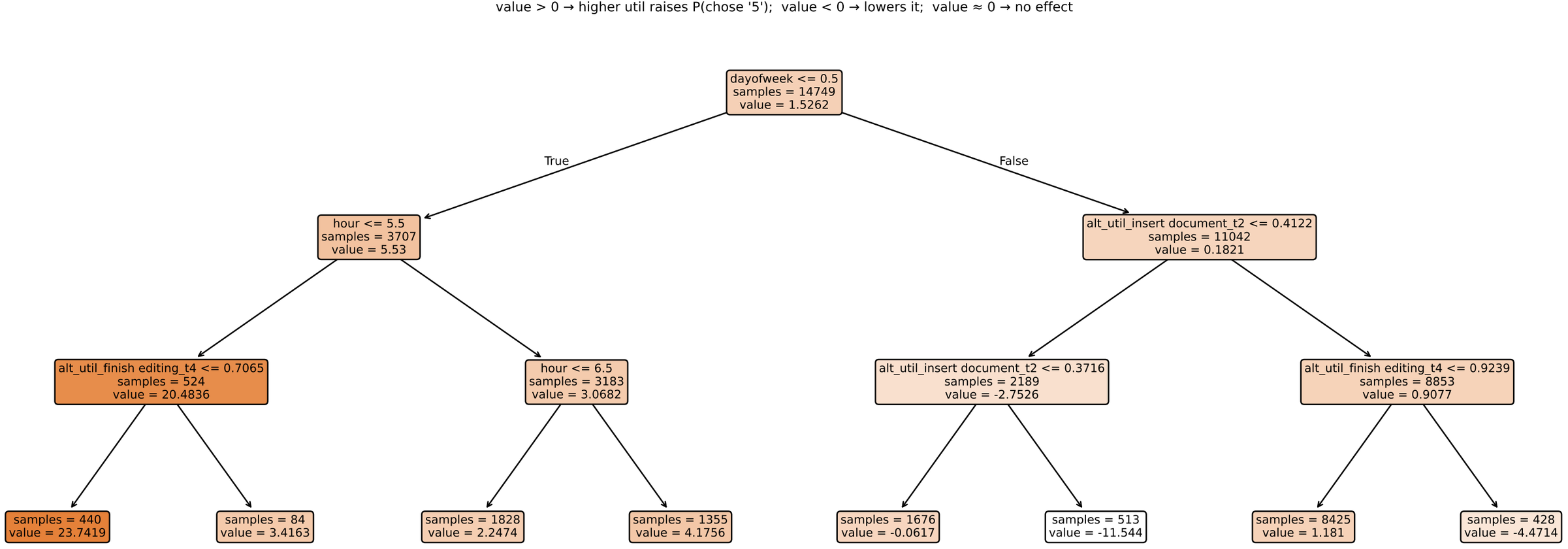

CATE summary tree | choice_set=(2, 3, 4, 5) | transition='4' | top feature: alt_util_insert document_t2
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_insert document_t2 <= 0.9122
samples = 14749
value = -0.7308
True
False
alt_util_insert document_t2 <= 0.8986
samples = 13896
value = -0.418
dayofweek <= 3.5
samples = 853
value = -5.826
hour <= 11.5
samples = 12943
value = -0.2801
hour <= 6.5
samples = 953
value = -2.2916
dayofweek <= 2.5
samples = 743
value = -5.1813
hour <= 8.5
samples = 110
value = -10.1806
samples = 11288
value = -0.1347
samples = 1655
value = -1.2715
samples = 305
value = -4.1914
samples = 648
value = -1.3973
samples = 483
value = -4.5169
samples = 260
value = -6.4154
samples = 44
value = -12.6696
samples = 66
value = -8.5213

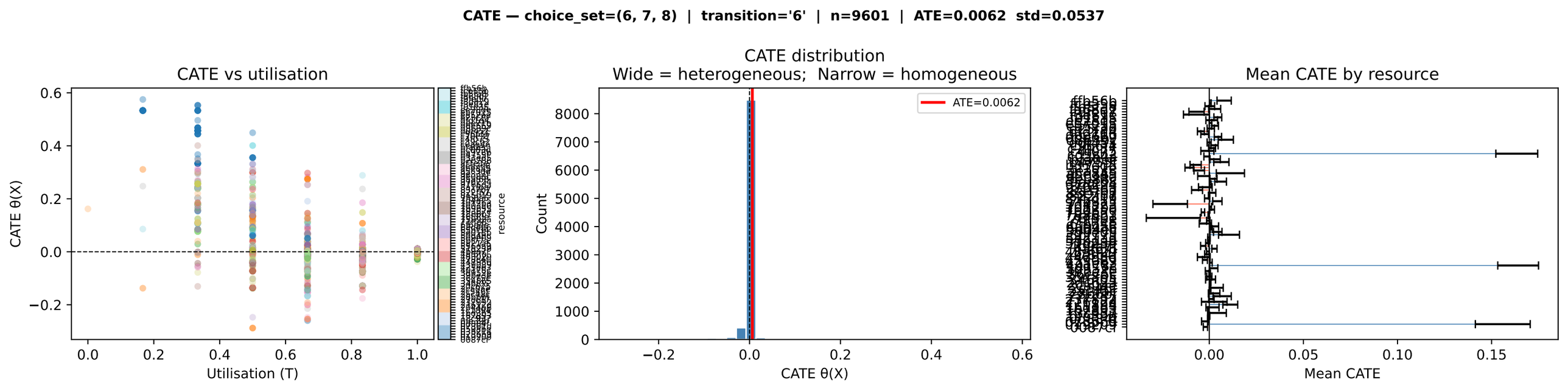
CATE — choice_set=(2, 3, 4, 5) | transition='5' | n=14749 | ATE=1.5262 std=5.6350

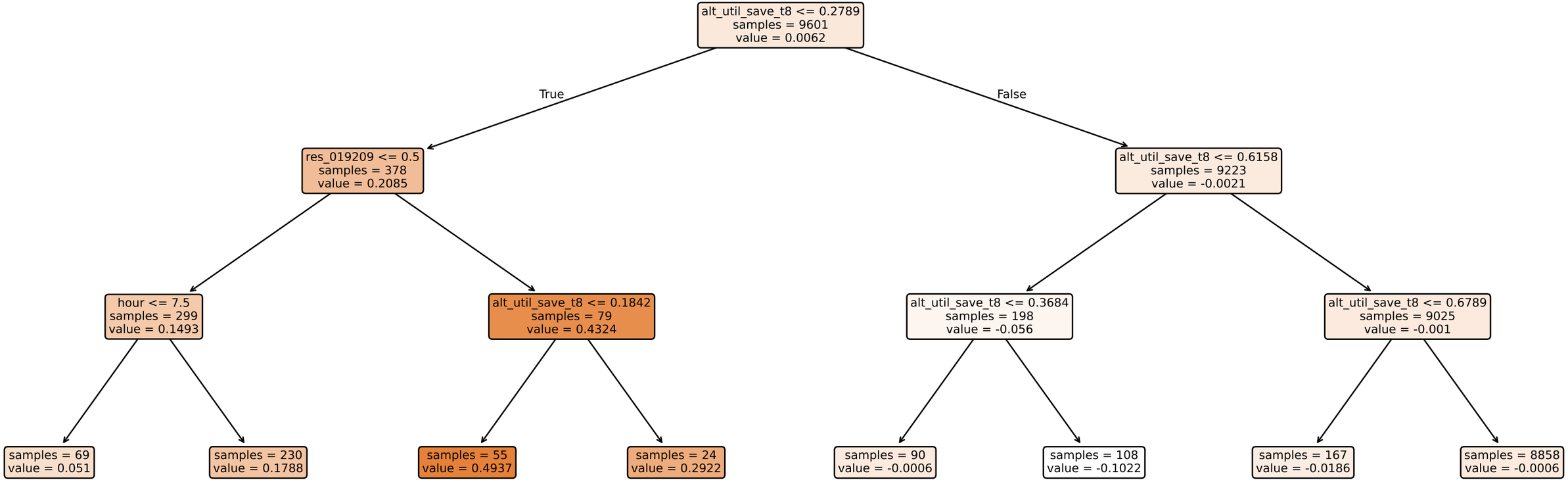


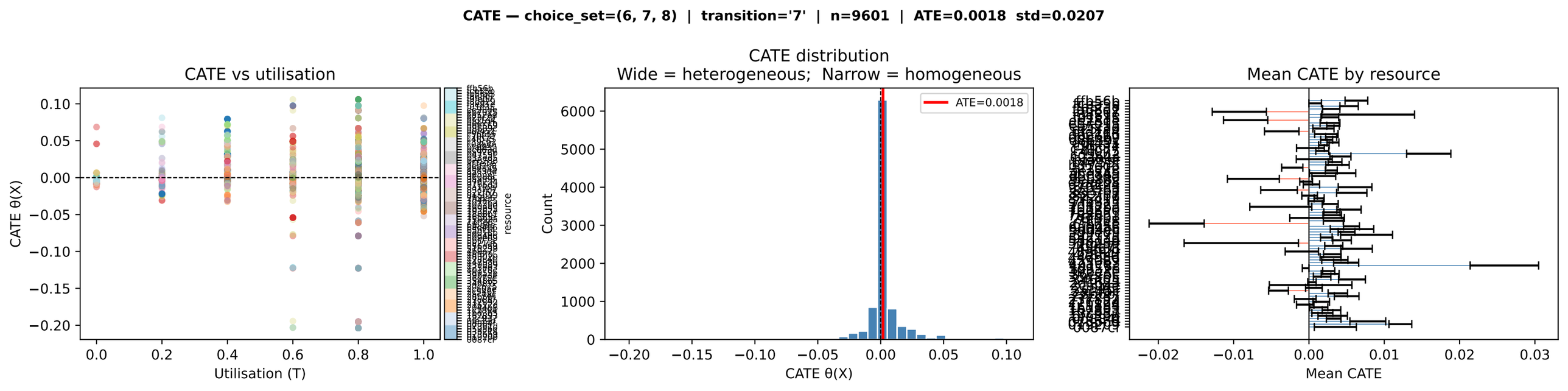


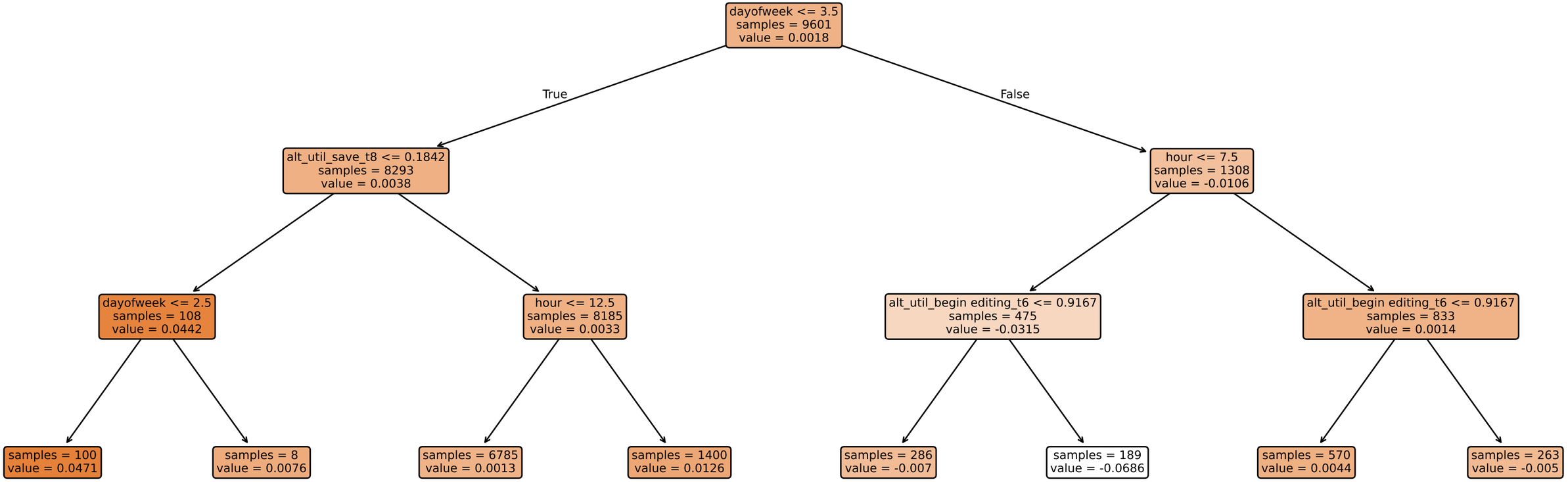


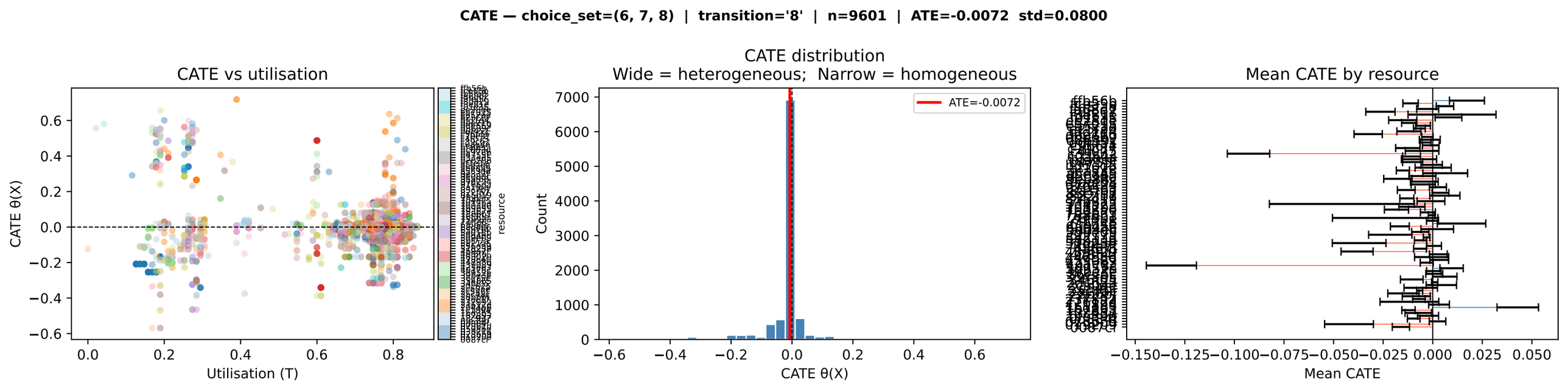

CATE summary tree | choice_set=(2, 3, 4, 5) | transition='5' | top feature: hour
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 0.5
samples = 14749
value = 1.5262
True
False
hour <= 5.5
samples = 3707
value = 5.53
alt_util_insert document_t2 <= 0.4122
samples = 11042
value = 0.1821
alt_util_finish editing_t4 <= 0.7065
samples = 524
value = 20.4836
hour <= 6.5
samples = 3183
value = 3.0682
alt_util_insert document_t2 <= 0.3716
samples = 2189
value = -2.7526
alt_util_finish editing_t4 <= 0.9239
samples = 8853
value = 0.9077
samples = 440
value = 23.7419
samples = 84
value = 3.4163
samples = 1828
value = 2.2474
samples = 1355
value = 4.1756
samples = 1676
value = -0.0617
samples = 513
value = -11.544
samples = 8425
value = 1.181
samples = 428
value = -4.4714

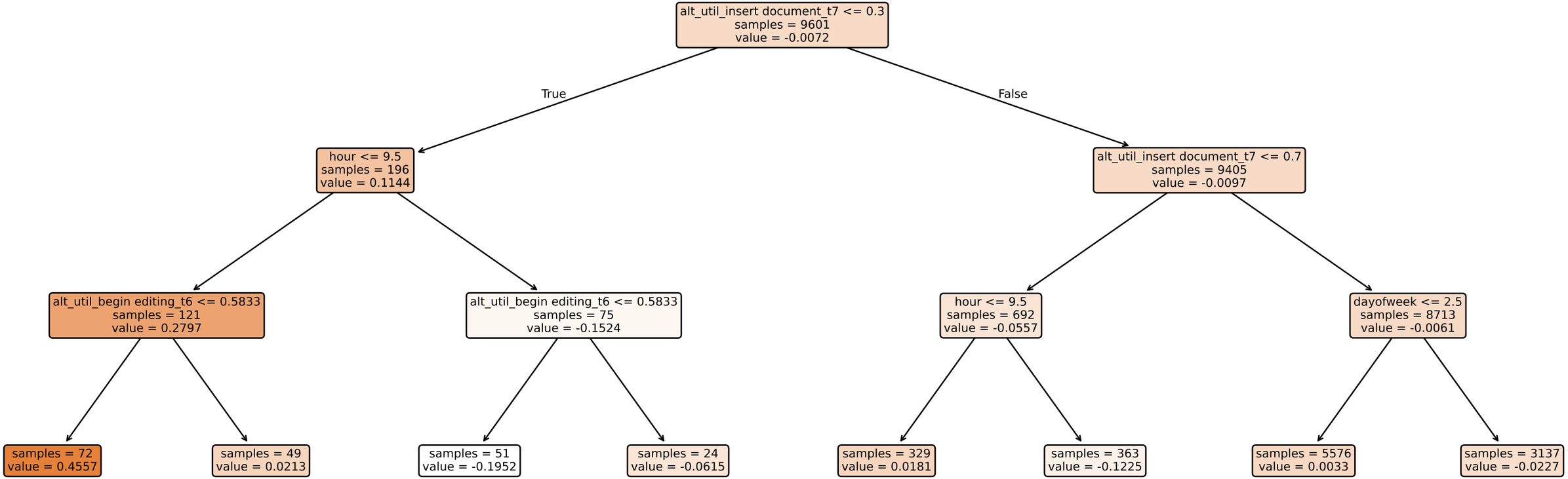

CATE — choice_set=(6, 7, 8) | transition='6' | n=9601 | ATE=0.0062 std=0.0537
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0062
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(6, 7, 8) | transition='6' | top feature: alt_util_save_t8
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t8 <= 0.2789
samples = 9601
value = 0.0062
True
False
res_019209 <= 0.5
samples = 378
value = 0.2085
alt_util_save_t8 <= 0.6158
samples = 9223
value = -0.0021
hour <= 7.5
samples = 299
value = 0.1493
alt_util_save_t8 <= 0.1842
samples = 79
value = 0.4324
alt_util_save_t8 <= 0.3684
samples = 198
value = -0.056
alt_util_save_t8 <= 0.6789
samples = 9025
value = -0.001
samples = 69
value = 0.051
samples = 230
value = 0.1788
samples = 55
value = 0.4937
samples = 24
value = 0.2922
samples = 90
value = -0.0006
samples = 108
value = -0.1022
samples = 167
value = -0.0186
samples = 8858
value = -0.0006

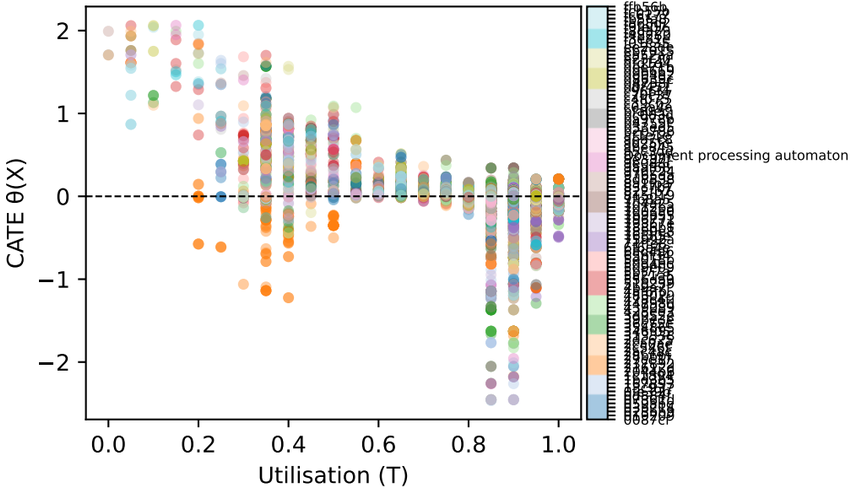

CATE — choice_set=(6, 7, 8) | transition='7' | n=9601 | ATE=0.0018 std=0.0207
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0018
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

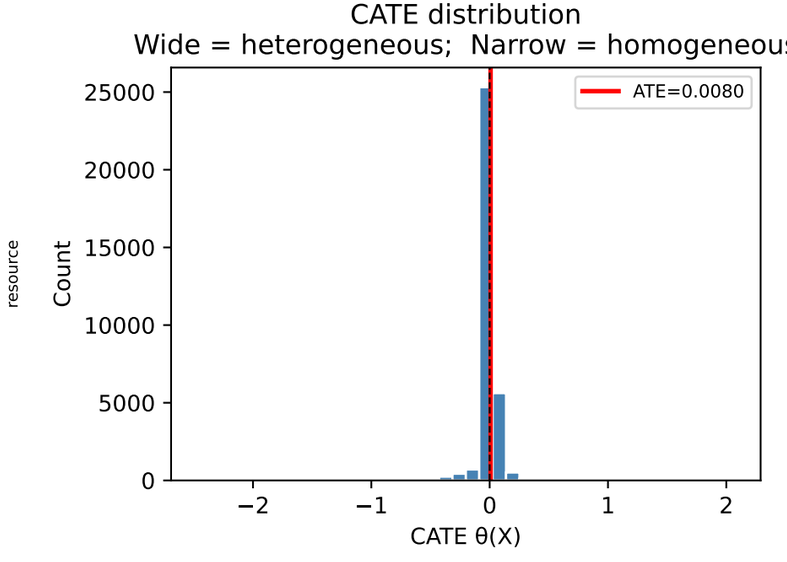

CATE summary tree | choice_set=(6, 7, 8) | transition='7' | top feature: hour
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 3.5
samples = 9601
value = 0.0018
True
False
alt_util_save_t8 <= 0.1842
samples = 8293
value = 0.0038
hour <= 7.5
samples = 1308
value = -0.0106
dayofweek <= 2.5
samples = 108
value = 0.0442
hour <= 12.5
samples = 8185
value = 0.0033
alt_util_begin editing_t6 <= 0.9167
samples = 475
value = -0.0315
alt_util_begin editing_t6 <= 0.9167
samples = 833
value = 0.0014
samples = 100
value = 0.0471
samples = 8
value = 0.0076
samples = 6785
value = 0.0013
samples = 1400
value = 0.0126
samples = 286
value = -0.007
samples = 189
value = -0.0686
samples = 570
value = 0.0044
samples = 263
value = -0.005

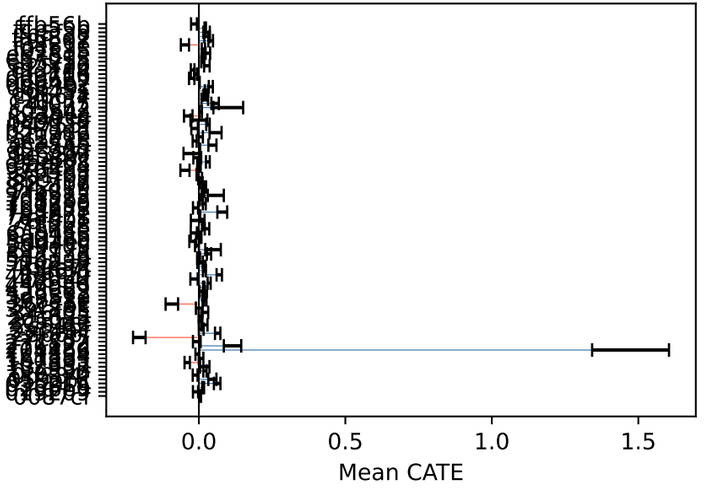

CATE — choice_set=(6, 7, 8) | transition='8' | n=9601 | ATE=-0.0072 std=0.0800
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-0.0072
Mean CATE by resource
Mean CATE

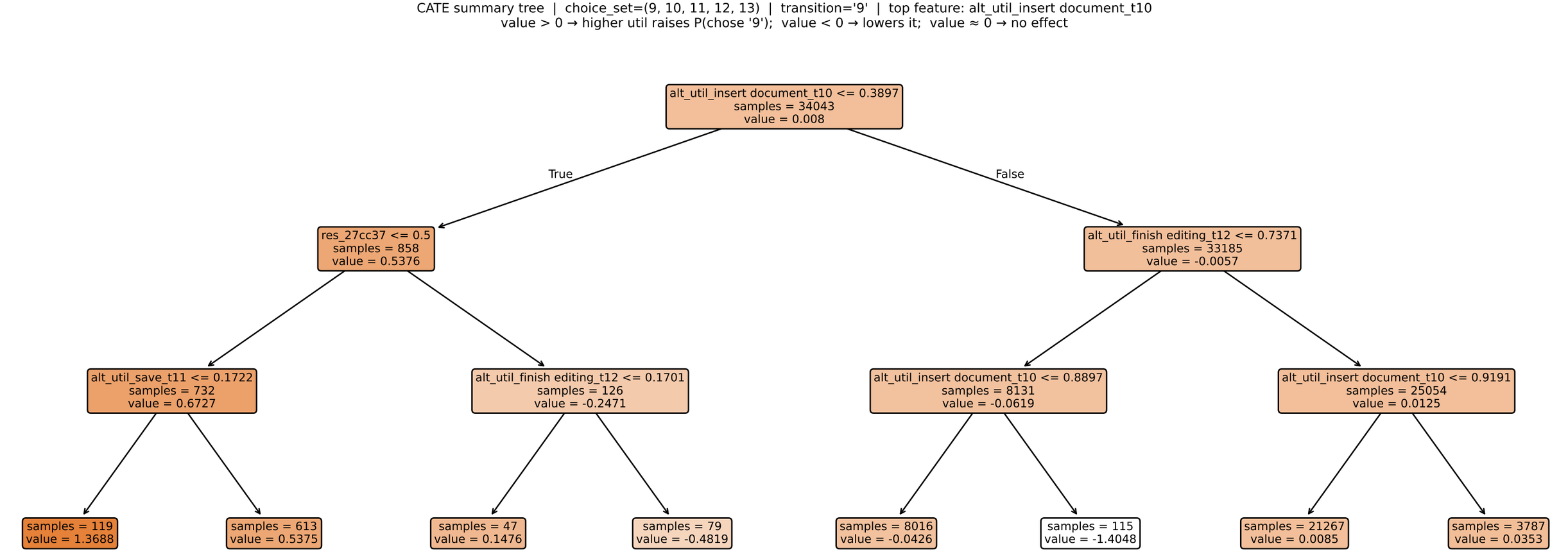

CATE summary tree | choice_set=(6, 7, 8) | transition='8' | top feature: hour
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_insert document_t7 <= 0.3
samples = 9601
value = -0.0072
True
False
hour <= 9.5
samples = 196
value = 0.1144
alt_util_insert document_t7 <= 0.7
samples = 9405
value = -0.0097
alt_util_begin editing_t6 <= 0.5833
samples = 121
value = 0.2797
alt_util_begin editing_t6 <= 0.5833
samples = 75
value = -0.1524
hour <= 9.5
samples = 692
value = -0.0557
dayofweek <= 2.5
samples = 8713
value = -0.0061
samples = 72
value = 0.4557
samples = 49
value = 0.0213
samples = 51
value = -0.1952
samples = 24
value = -0.0615
samples = 329
value = 0.0181
samples = 363
value = -0.1225
samples = 5576
value = 0.0033
samples = 3137
value = -0.0227

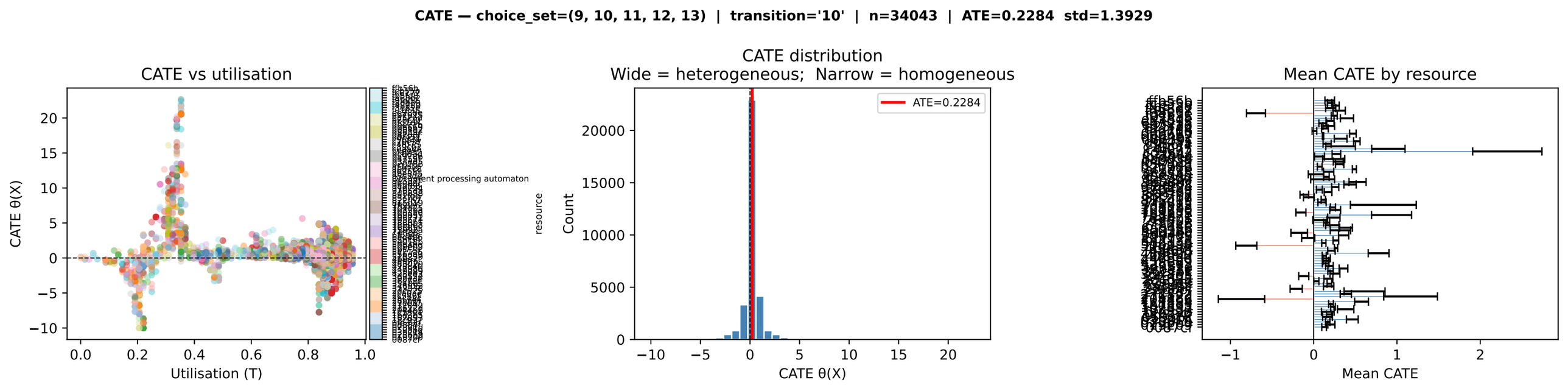
CATE — choice_set=(9, 10, 11, 12, 13) | transition='9' | n=34043 | ATE=0.0080 std=0.1815

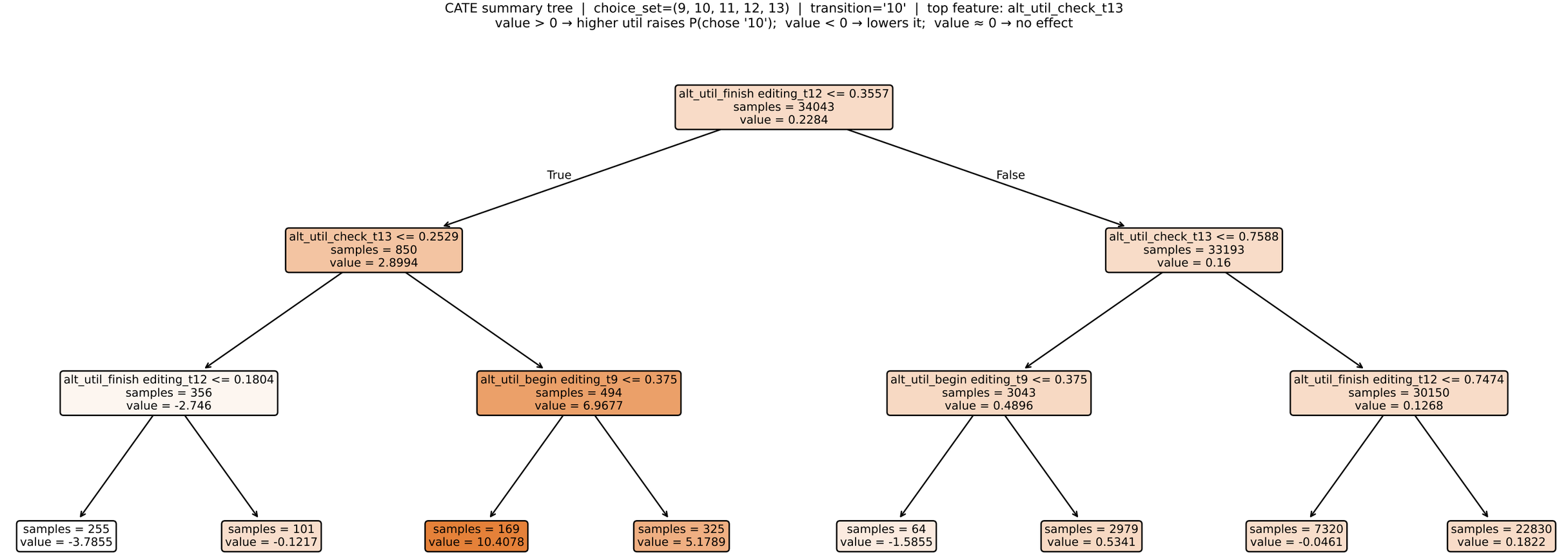


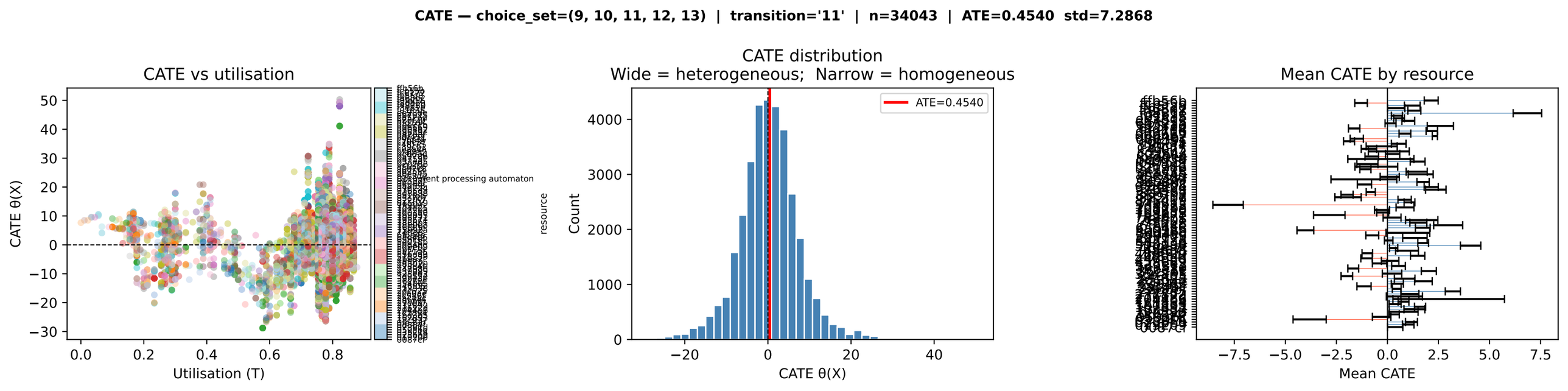


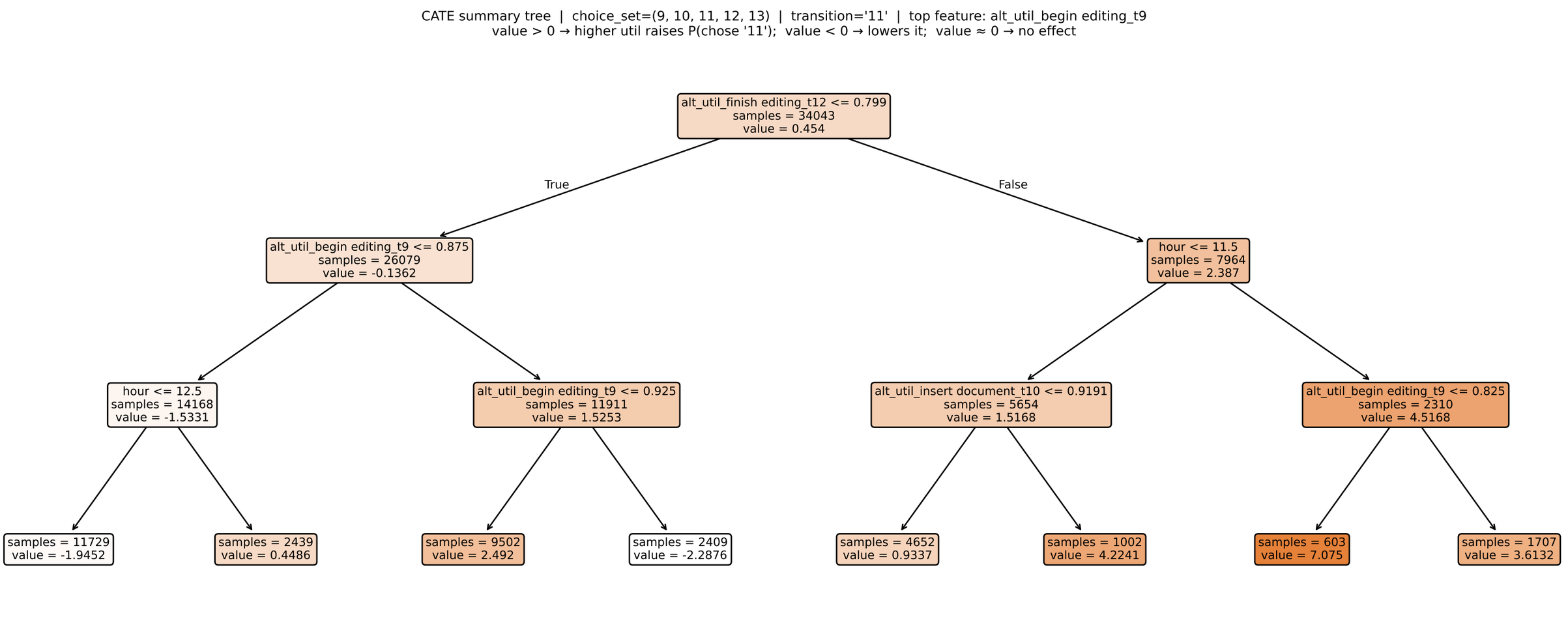


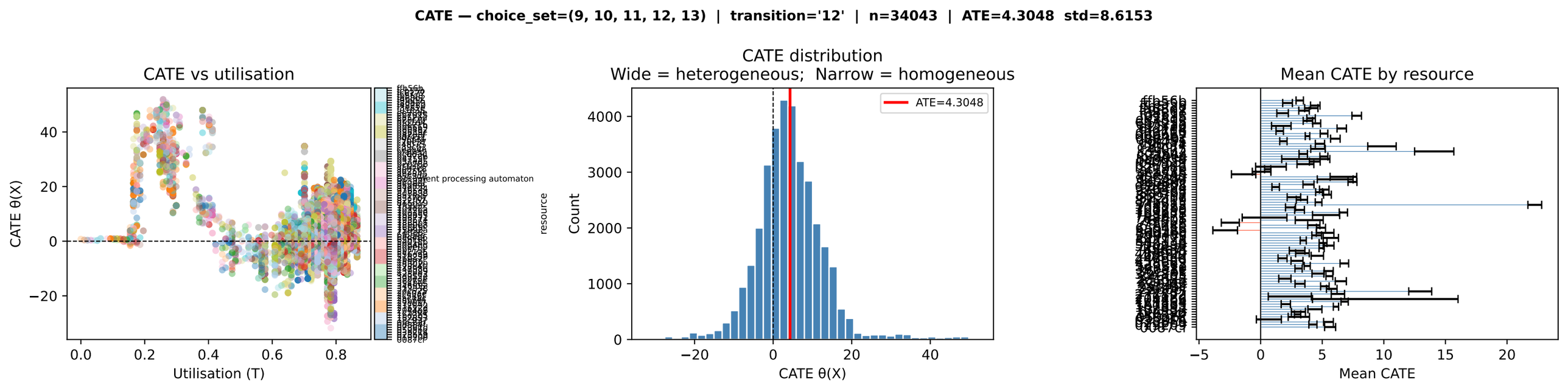

CATE summary tree | choice_set=(9, 10, 11, 12, 13) | transition='9' | top feature: alt_util_insert document_t10
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_insert document_t10 <= 0.3897
samples = 34043
value = 0.008
True
False
res_27cc37 <= 0.5
samples = 858
value = 0.5376
alt_util_finish editing_t12 <= 0.7371
samples = 33185
value = -0.0057
alt_util_save_t11 <= 0.1722
samples = 732
value = 0.6727
alt_util_finish editing_t12 <= 0.1701
samples = 126
value = -0.2471
alt_util_insert document_t10 <= 0.8897
samples = 8131
value = -0.0619
alt_util_insert document_t10 <= 0.9191
samples = 25054
value = 0.0125
samples = 119
value = 1.3688
samples = 613
value = 0.5375
samples = 47
value = 0.1476
samples = 79
value = -0.4819
samples = 8016
value = -0.0426
samples = 115
value = -1.4048
samples = 21267
value = 0.0085
samples = 3787
value = 0.0353

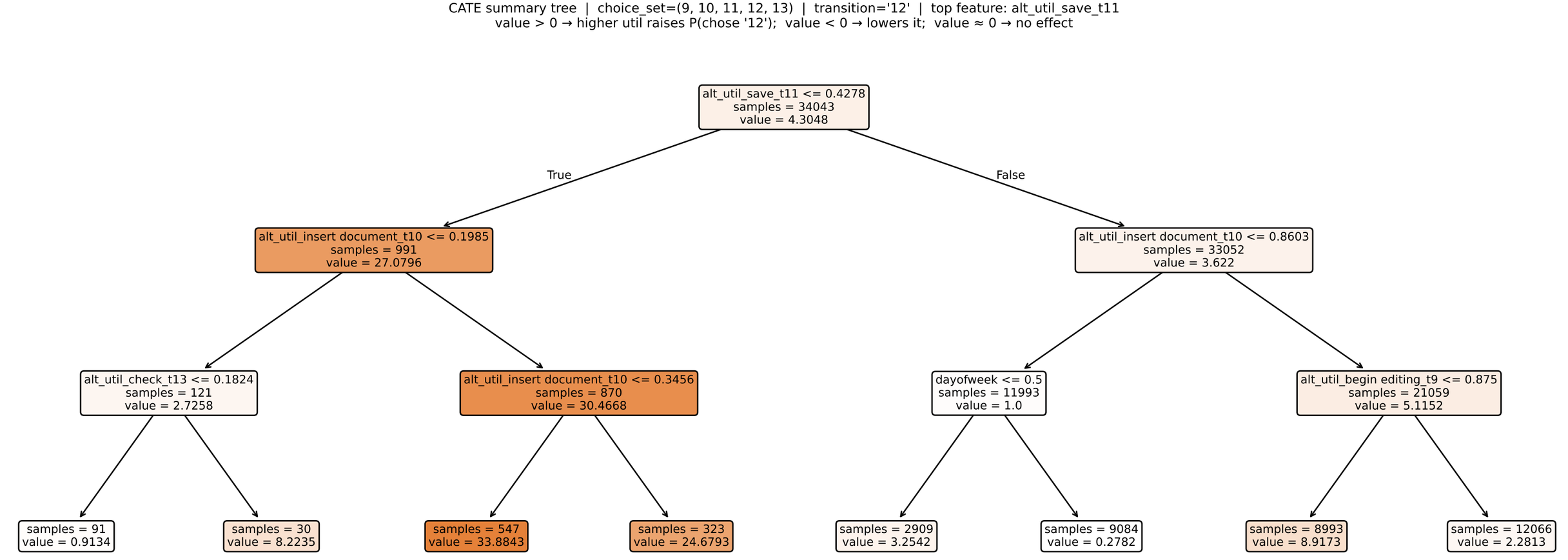

CATE — choice_set=(9, 10, 11, 12, 13) | transition='10' | n=34043 | ATE=0.2284 std=1.3929
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.2284
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

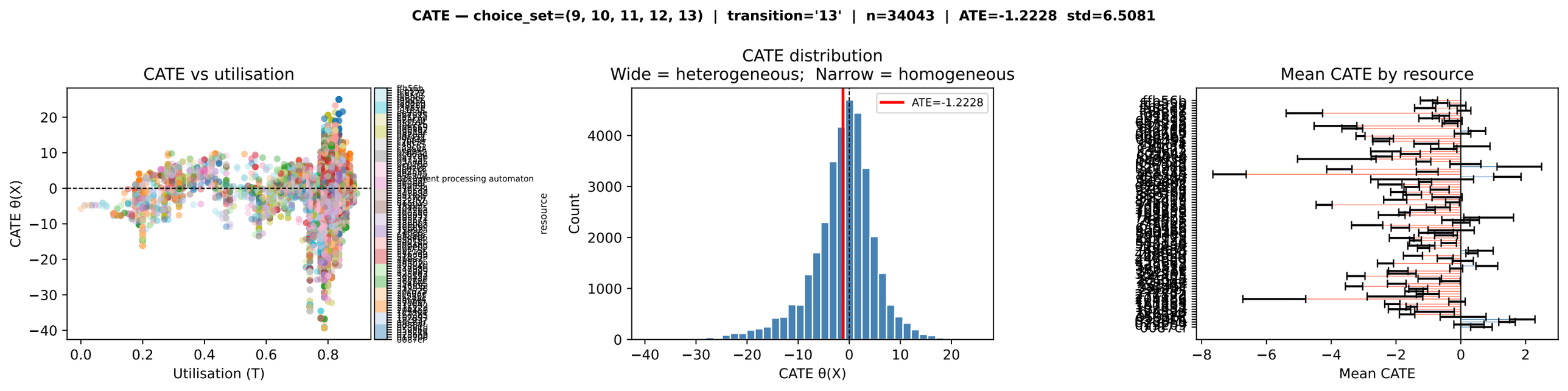

CATE summary tree | choice_set=(9, 10, 11, 12, 13) | transition='10' | top feature: alt_util_check_t13
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t12 <= 0.3557
samples = 34043
value = 0.2284
True
False
alt_util_check_t13 <= 0.2529
samples = 850
value = 2.8994
alt_util_check_t13 <= 0.7588
samples = 33193
value = 0.16
alt_util_finish editing_t12 <= 0.1804
samples = 356
value = -2.746
alt_util_begin editing_t9 <= 0.375
samples = 494
value = 6.9677
alt_util_begin editing_t9 <= 0.375
samples = 3043
value = 0.4896
alt_util_finish editing_t12 <= 0.7474
samples = 30150
value = 0.1268
samples = 255
value = -3.7855
samples = 101
value = -0.1217
samples = 169
value = 10.4078
samples = 325
value = 5.1789
samples = 64
value = -1.5855
samples = 2979
value = 0.5341
samples = 7320
value = -0.0461
samples = 22830
value = 0.1822

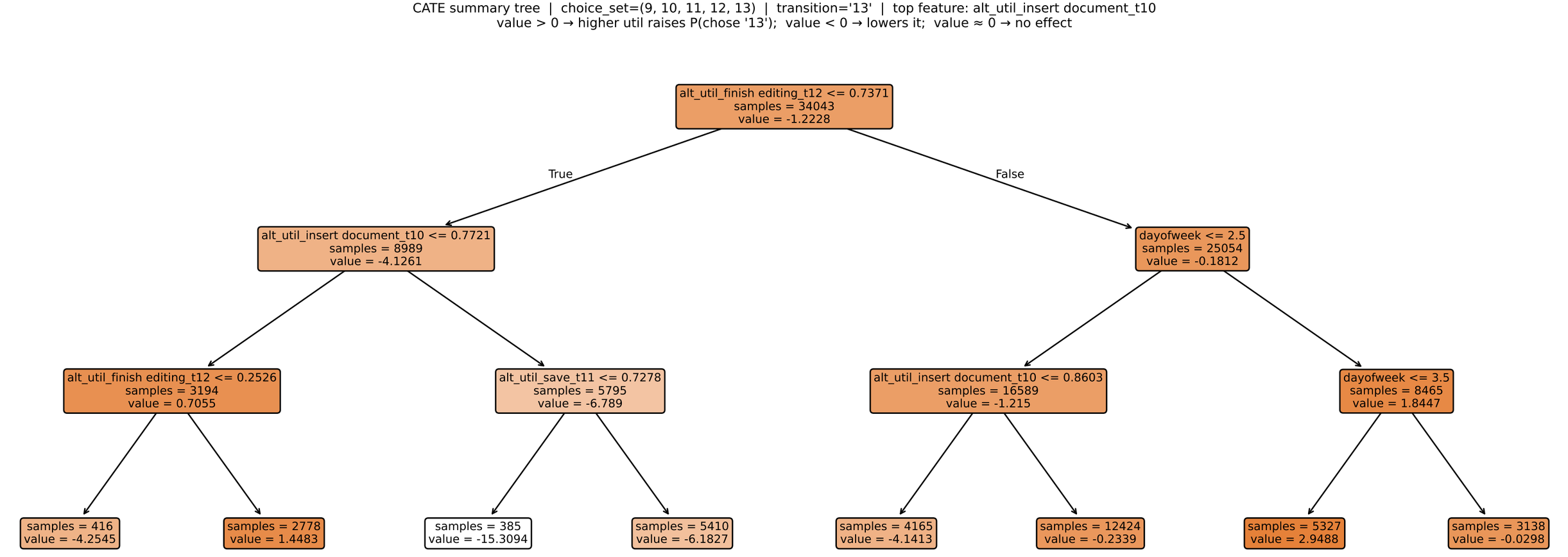

CATE — choice_set=(9, 10, 11, 12, 13) | transition='11' | n=34043 | ATE=0.4540 std=7.2868
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4540
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

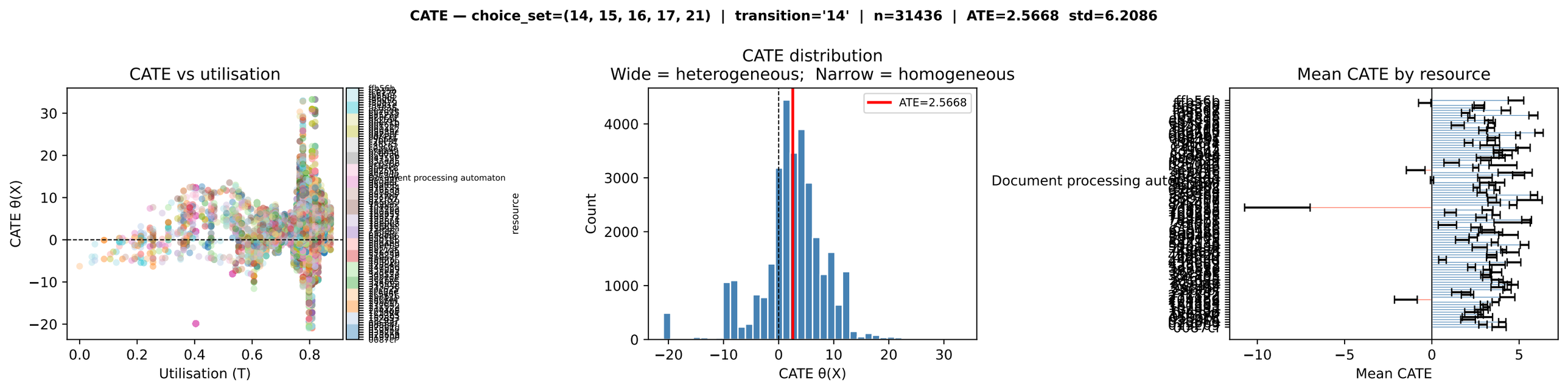

CATE summary tree | choice_set=(9, 10, 11, 12, 13) | transition='11' | top feature: alt_util_begin editing_t9
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t12 <= 0.799
samples = 34043
value = 0.454
True
False
alt_util_begin editing_t9 <= 0.875
samples = 26079
value = -0.1362
hour <= 11.5
samples = 7964
value = 2.387
hour <= 12.5
samples = 14168
value = -1.5331
alt_util_begin editing_t9 <= 0.925
samples = 11911
value = 1.5253
alt_util_insert document_t10 <= 0.9191
samples = 5654
value = 1.5168
alt_util_begin editing_t9 <= 0.825
samples = 2310
value = 4.5168
samples = 11729
value = -1.9452
samples = 2439
value = 0.4486
samples = 9502
value = 2.492
samples = 2409
value = -2.2876
samples = 4652
value = 0.9337
samples = 1002
value = 4.2241
samples = 603
value = 7.075
samples = 1707
value = 3.6132

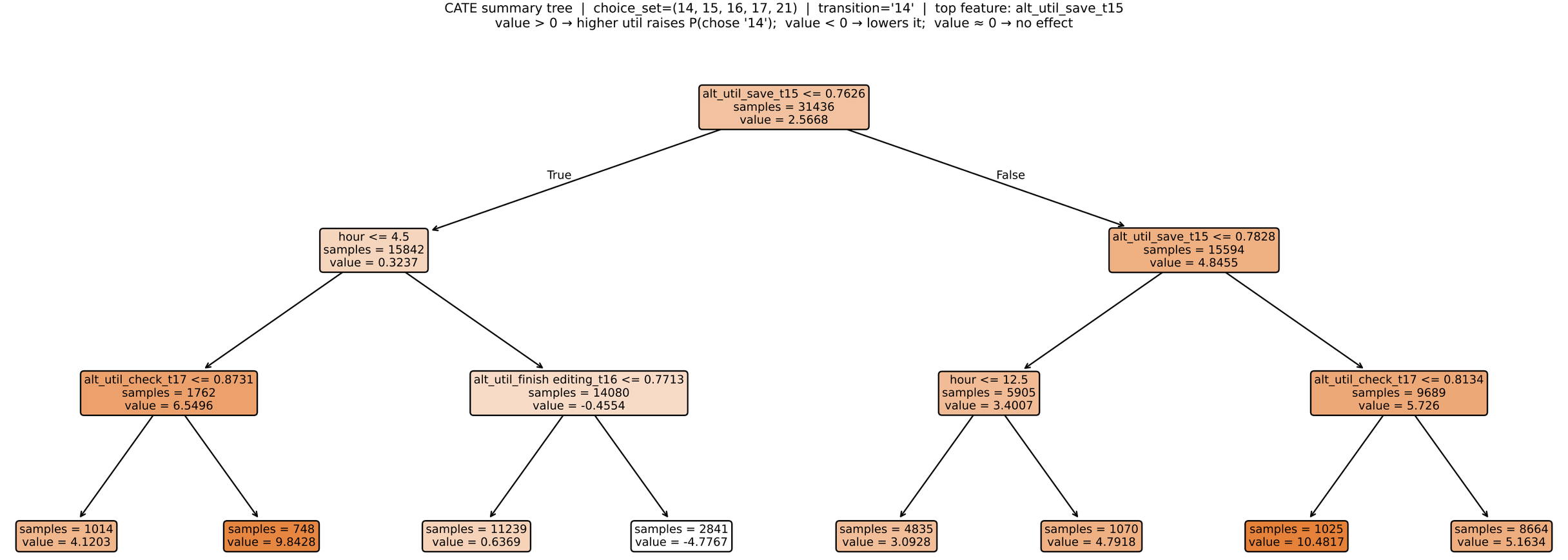

CATE — choice_set=(9, 10, 11, 12, 13) | transition='12' | n=34043 | ATE=4.3048 std=8.6153
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=4.3048
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(9, 10, 11, 12, 13) | transition='12' | top feature: alt_util_save_t11
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t11 <= 0.4278
samples = 34043
value = 4.3048
True
False
alt_util_insert document_t10 <= 0.1985
samples = 991
value = 27.0796
alt_util_insert document_t10 <= 0.8603
samples = 33052
value = 3.622
alt_util_check_t13 <= 0.1824
samples = 121
value = 2.7258
alt_util_insert document_t10 <= 0.3456
samples = 870
value = 30.4668
dayofweek <= 0.5
samples = 11993
value = 1.0
alt_util_begin editing_t9 <= 0.875
samples = 21059
value = 5.1152
samples = 91
value = 0.9134
samples = 30
value = 8.2235
samples = 547
value = 33.8843
samples = 323
value = 24.6793
samples = 2909
value = 3.2542
samples = 9084
value = 0.2782
samples = 8993
value = 8.9173
samples = 12066
value = 2.2813

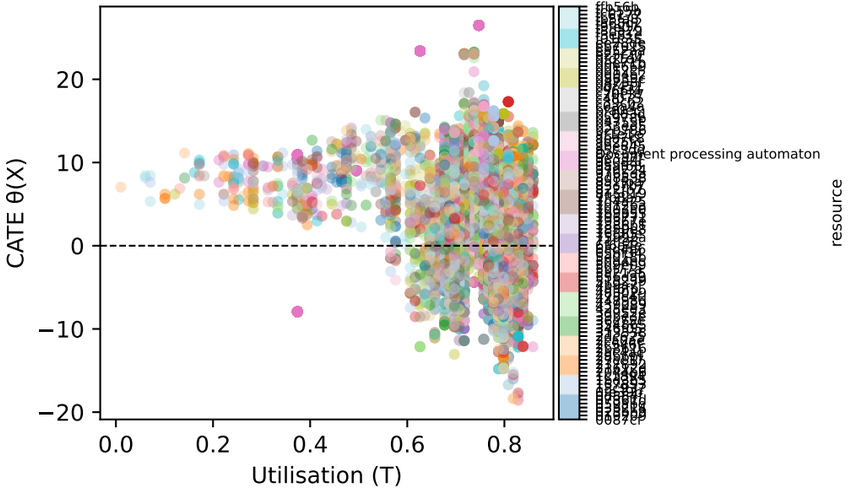

CATE — choice_set=(9, 10, 11, 12, 13) | transition='13' | n=34043 | ATE=-1.2228 std=6.5081
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.2228
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

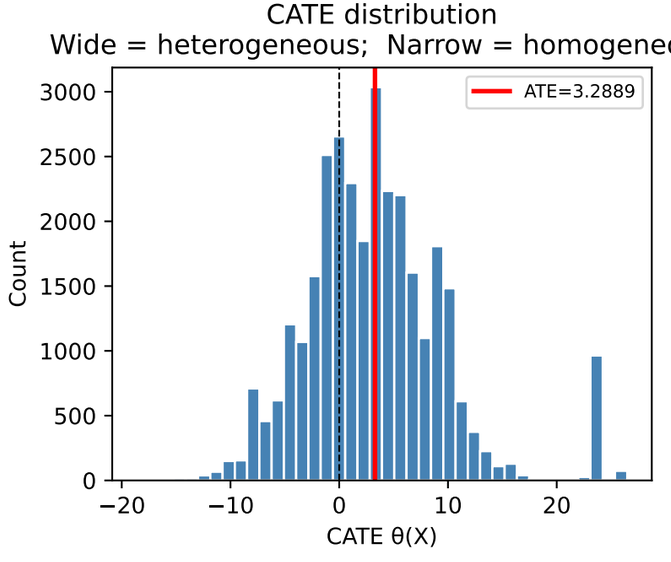

CATE summary tree | choice_set=(9, 10, 11, 12, 13) | transition='13' | top feature: alt_util_insert document_t10
value > 0 → higher util raises P(chose '13'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t12 <= 0.7371
samples = 34043
value = -1.2228
True
False
alt_util_insert document_t10 <= 0.7721
samples = 8989
value = -4.1261
dayofweek <= 2.5
samples = 25054
value = -0.1812
alt_util_finish editing_t12 <= 0.2526
samples = 3194
value = 0.7055
alt_util_save_t11 <= 0.7278
samples = 5795
value = -6.789
alt_util_insert document_t10 <= 0.8603
samples = 16589
value = -1.215
dayofweek <= 3.5
samples = 8465
value = 1.8447
samples = 416
value = -4.2545
samples = 2778
value = 1.4483
samples = 385
value = -15.3094
samples = 5410
value = -6.1827
samples = 4165
value = -4.1413
samples = 12424
value = -0.2339
samples = 5327
value = 2.9488
samples = 3138
value = -0.0298

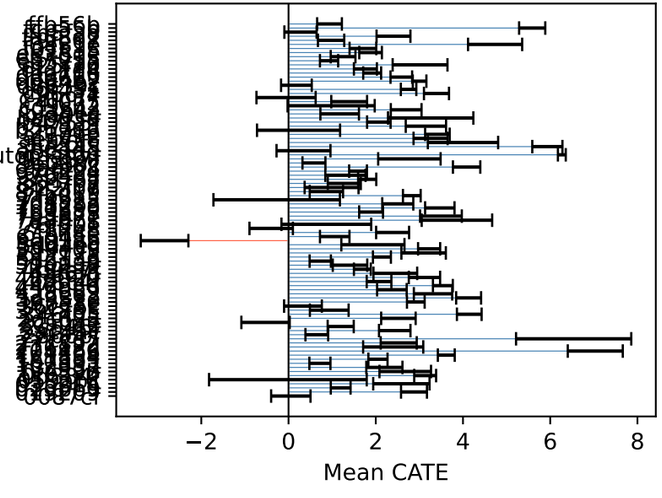

CATE — choice_set=(14, 15, 16, 17, 21) | transition='14' | n=31436 | ATE=2.5668 std=6.2086
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=2.5668
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

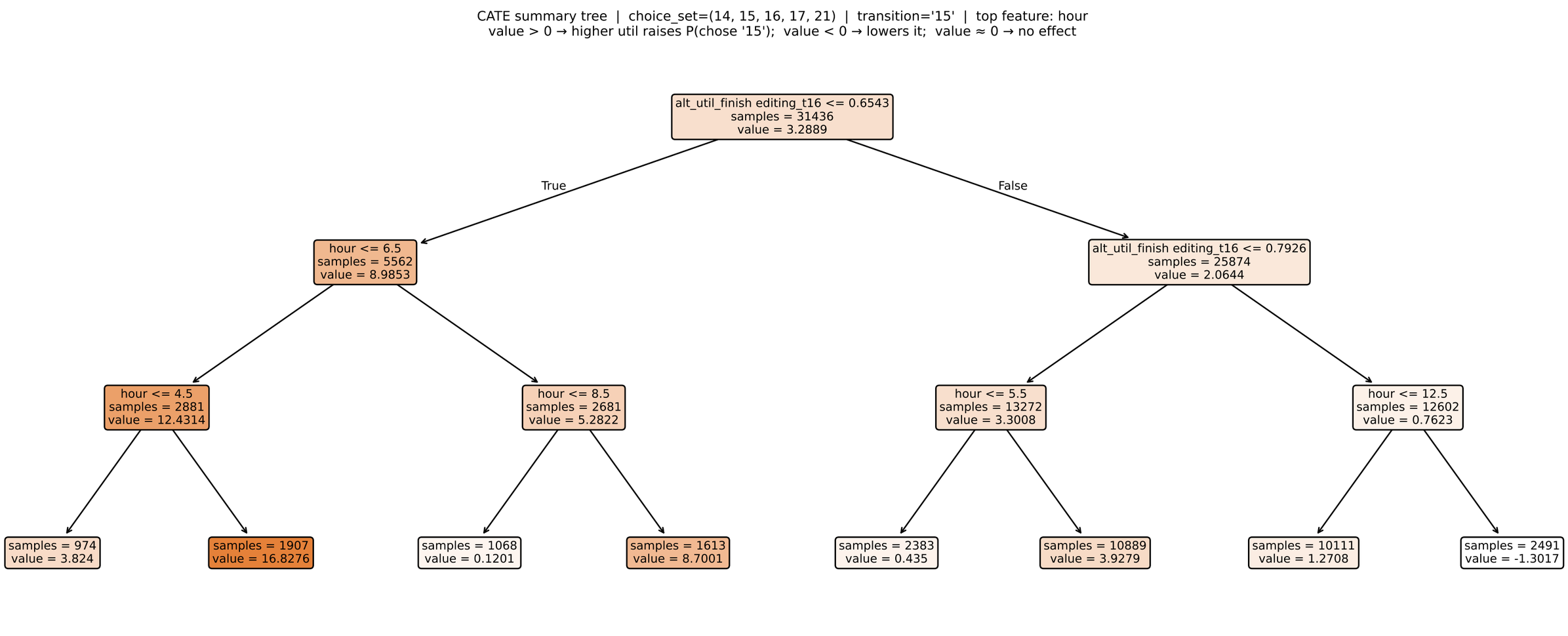

CATE summary tree | choice_set=(14, 15, 16, 17, 21) | transition='14' | top feature: alt_util_save_t15
value > 0 → higher util raises P(chose '14'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t15 <= 0.7626
samples = 31436
value = 2.5668
True
False
hour <= 4.5
samples = 15842
value = 0.3237
alt_util_save_t15 <= 0.7828
samples = 15594
value = 4.8455
alt_util_check_t17 <= 0.8731
samples = 1762
value = 6.5496
alt_util_finish editing_t16 <= 0.7713
samples = 14080
value = -0.4554
hour <= 12.5
samples = 5905
value = 3.4007
alt_util_check_t17 <= 0.8134
samples = 9689
value = 5.726
samples = 1014
value = 4.1203
samples = 748
value = 9.8428
samples = 11239
value = 0.6369
samples = 2841
value = -4.7767
samples = 4835
value = 3.0928
samples = 1070
value = 4.7918
samples = 1025
value = 10.4817
samples = 8664
value = 5.1634

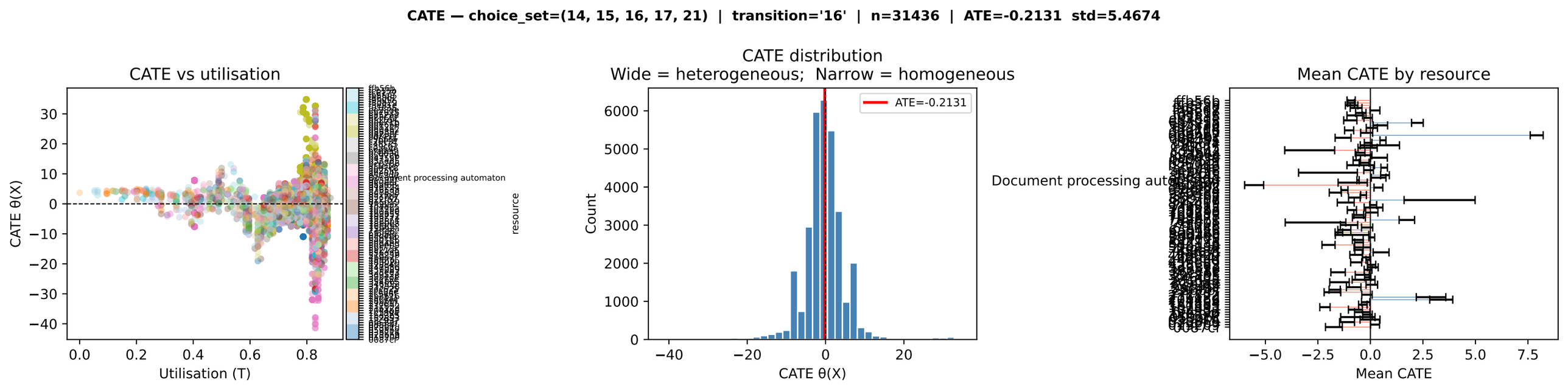
CATE — choice_set=(14, 15, 16, 17, 21) | transition='15' | n=31436 | ATE=3.2889 std=6.4138

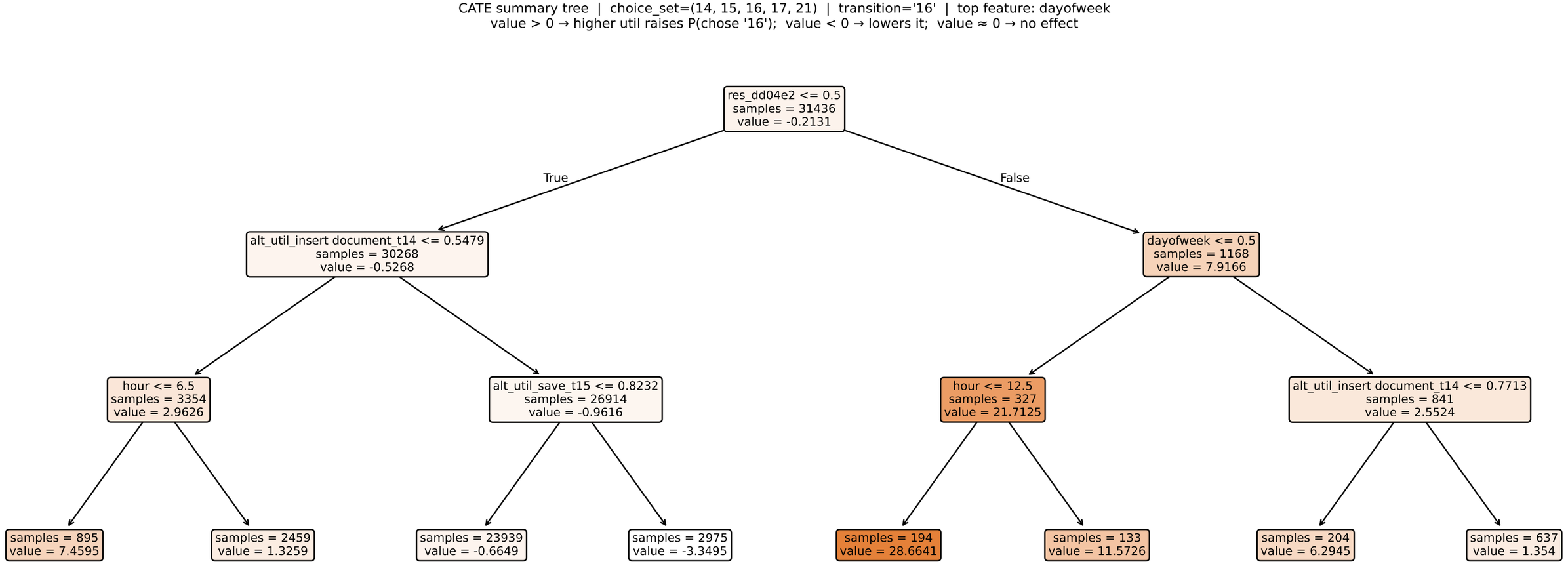




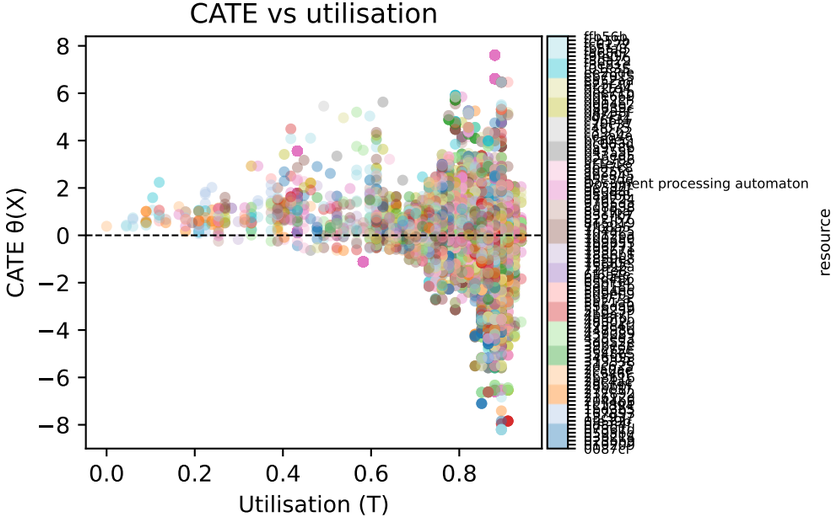


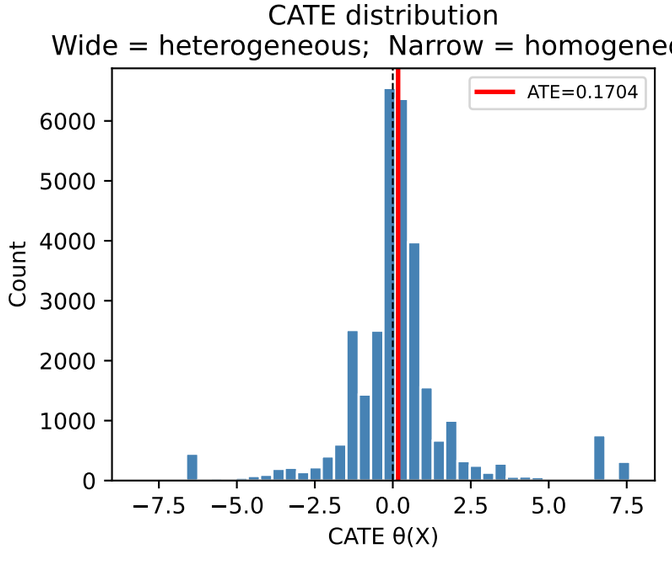

CATE summary tree | choice_set=(14, 15, 16, 17, 21) | transition='15' | top feature: hour
value > 0 → higher util raises P(chose '15'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t16 <= 0.6543
samples = 31436
value = 3.2889
True
False
hour <= 6.5
samples = 5562
value = 8.9853
alt_util_finish editing_t16 <= 0.7926
samples = 25874
value = 2.0644
hour <= 4.5
samples = 2881
value = 12.4314
hour <= 8.5
samples = 2681
value = 5.2822
hour <= 5.5
samples = 13272
value = 3.3008
hour <= 12.5
samples = 12602
value = 0.7623
samples = 974
value = 3.824
samples = 1907
value = 16.8276
samples = 1068
value = 0.1201
samples = 1613
value = 8.7001
samples = 2383
value = 0.435
samples = 10889
value = 3.9279
samples = 10111
value = 1.2708
samples = 2491
value = -1.3017

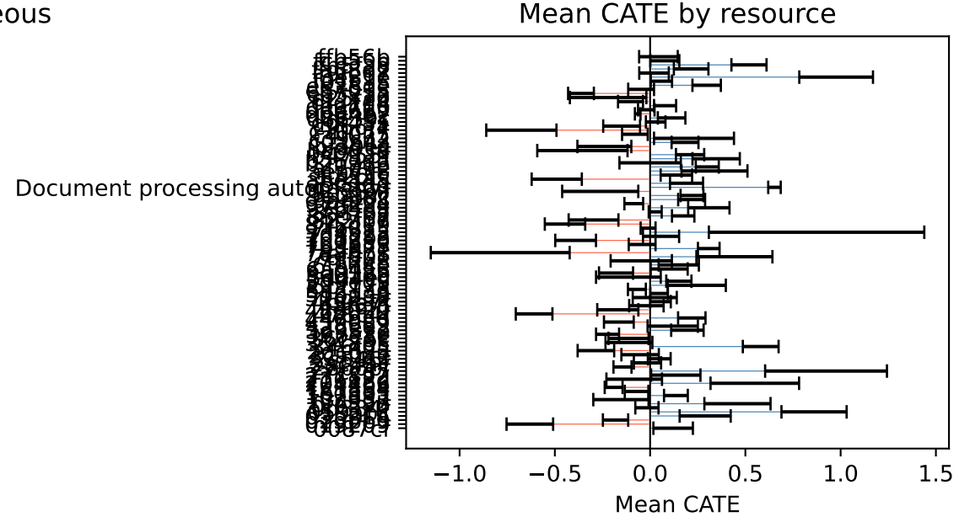

CATE — choice_set=(14, 15, 16, 17, 21) | transition='16' | n=31436 | ATE=-0.2131 std=5.4674
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.2131
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

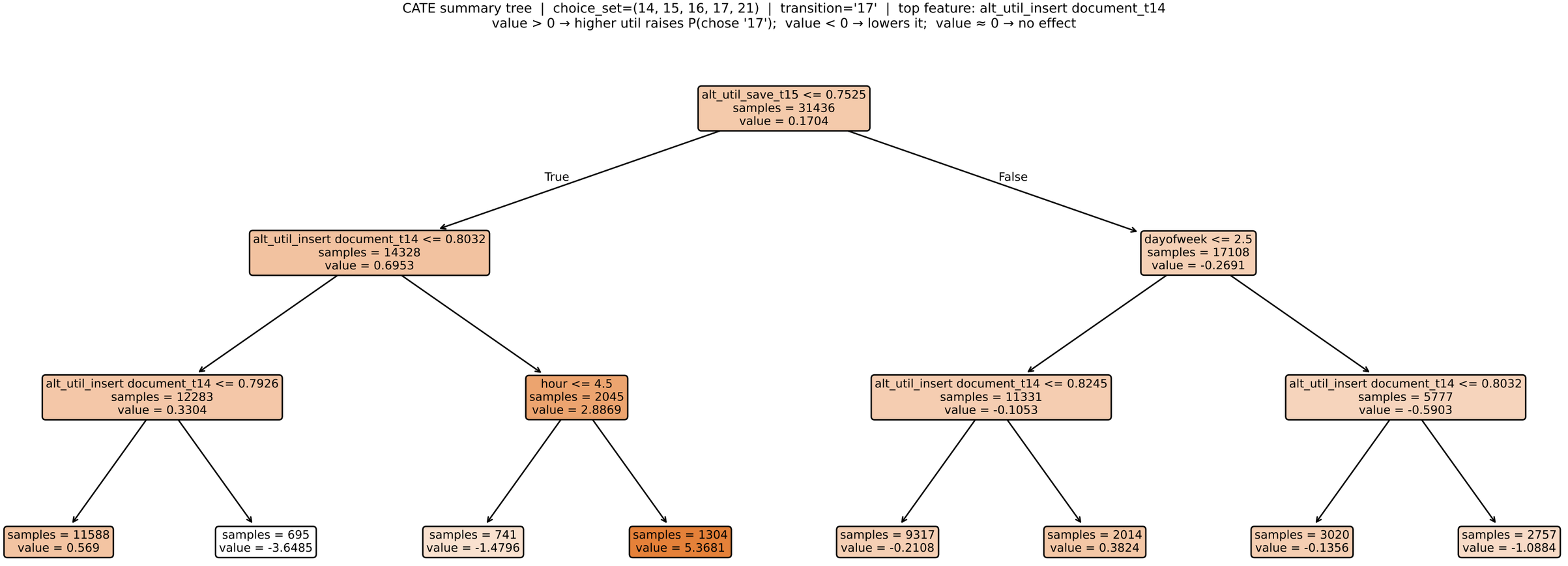

CATE summary tree | choice_set=(14, 15, 16, 17, 21) | transition='16' | top feature: dayofweek
value > 0 → higher util raises P(chose '16'); value < 0 → lowers it; value ≈ 0 → no effect
res_dd04e2 <= 0.5
samples = 31436
value = -0.2131
True
False
alt_util_insert document_t14 <= 0.5479
samples = 30268
value = -0.5268
dayofweek <= 0.5
samples = 1168
value = 7.9166
hour <= 6.5
samples = 3354
value = 2.9626
alt_util_save_t15 <= 0.8232
samples = 26914
value = -0.9616
hour <= 12.5
samples = 327
value = 21.7125
alt_util_insert document_t14 <= 0.7713
samples = 841
value = 2.5524
samples = 895
value = 7.4595
samples = 2459
value = 1.3259
samples = 23939
value = -0.6649
samples = 2975
value = -3.3495
samples = 194
value = 28.6641
samples = 133
value = 11.5726
samples = 204
value = 6.2945
samples = 637
value = 1.354

CATE — choice_set=(14, 15, 16, 17, 21) | transition='17' | n=31436 | ATE=0.1704 std=1.9322

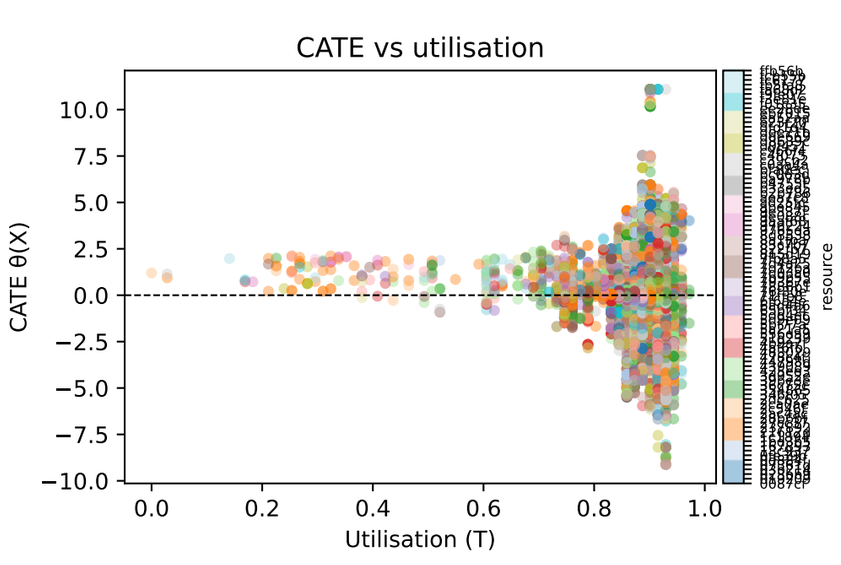


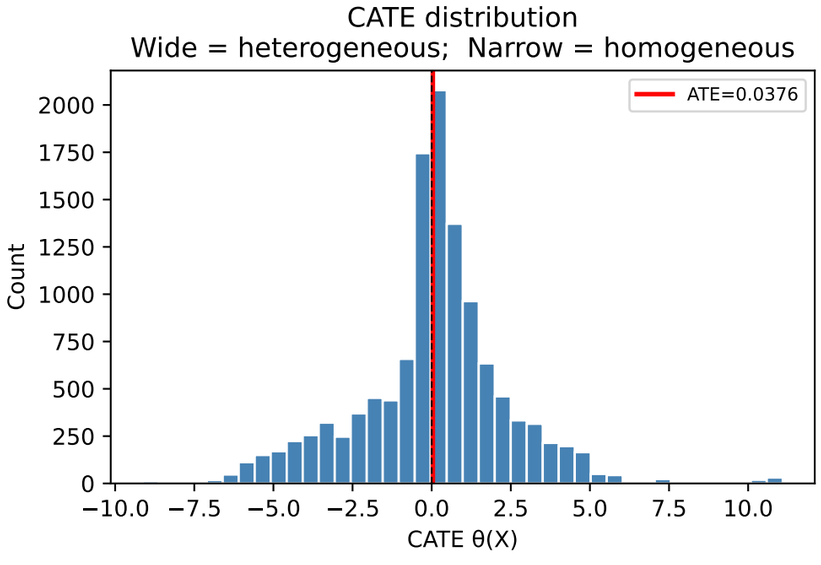


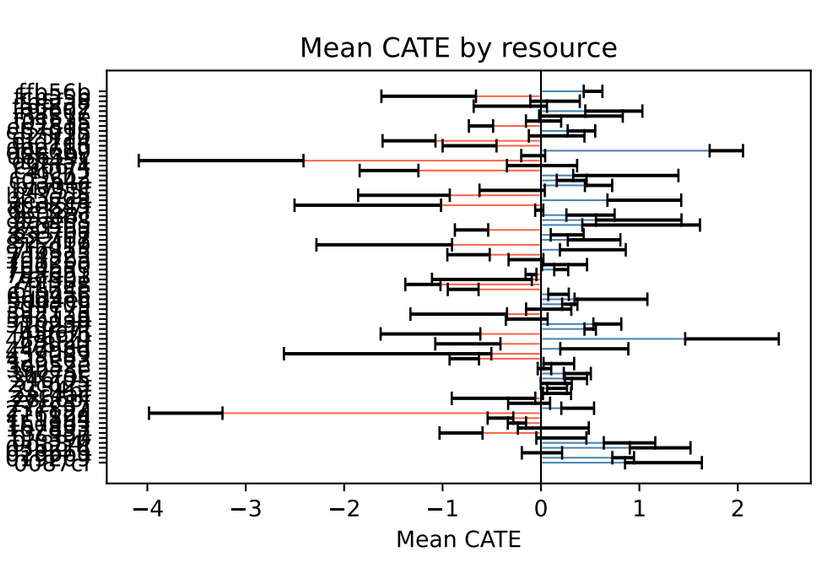


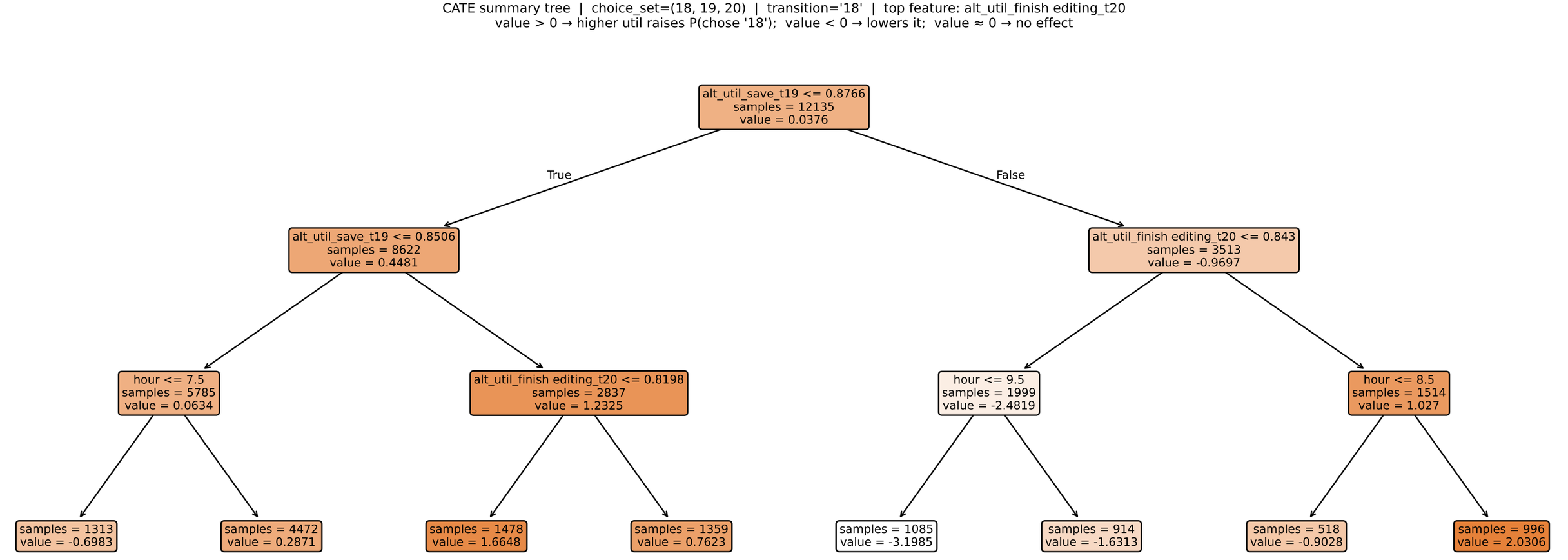

CATE summary tree | choice_set=(14, 15, 16, 17, 21) | transition='17' | top feature: alt_util_insert document_t14
value > 0 → higher util raises P(chose '17'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t15 <= 0.7525
samples = 31436
value = 0.1704
True
False
alt_util_insert document_t14 <= 0.8032
samples = 14328
value = 0.6953
dayofweek <= 2.5
samples = 17108
value = -0.2691
alt_util_insert document_t14 <= 0.7926
samples = 12283
value = 0.3304
hour <= 4.5
samples = 2045
value = 2.8869
alt_util_insert document_t14 <= 0.8245
samples = 11331
value = -0.1053
alt_util_insert document_t14 <= 0.8032
samples = 5777
value = -0.5903
samples = 11588
value = 0.569
samples = 695
value = -3.6485
samples = 741
value = -1.4796
samples = 1304
value = 5.3681
samples = 9317
value = -0.2108
samples = 2014
value = 0.3824
samples = 3020
value = -0.1356
samples = 2757
value = -1.0884

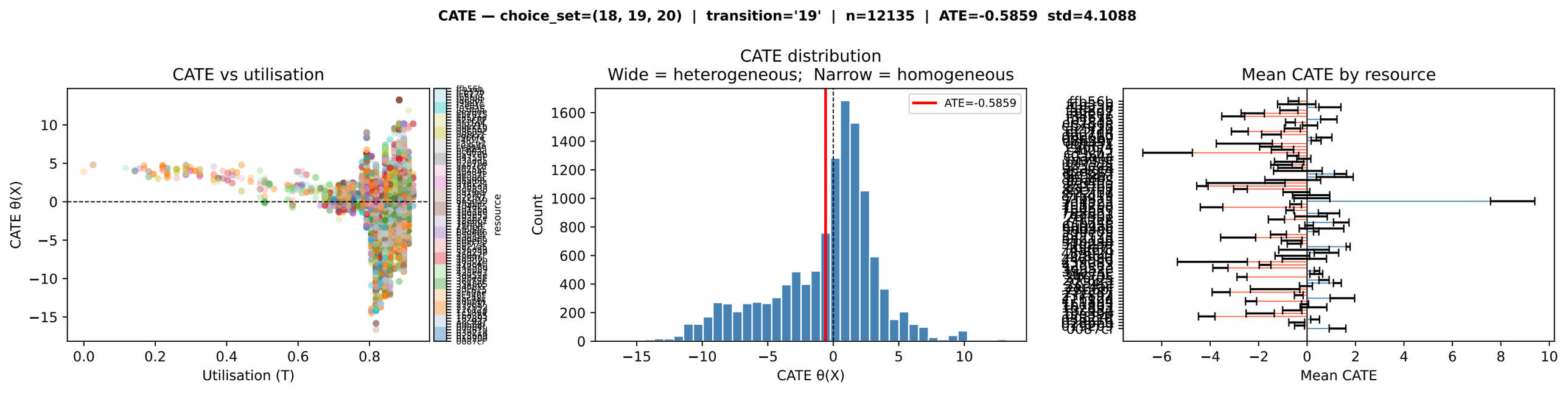
CATE — choice_set=(18, 19, 20) | transition='18' | n=12135 | ATE=0.0376 std=2.3555

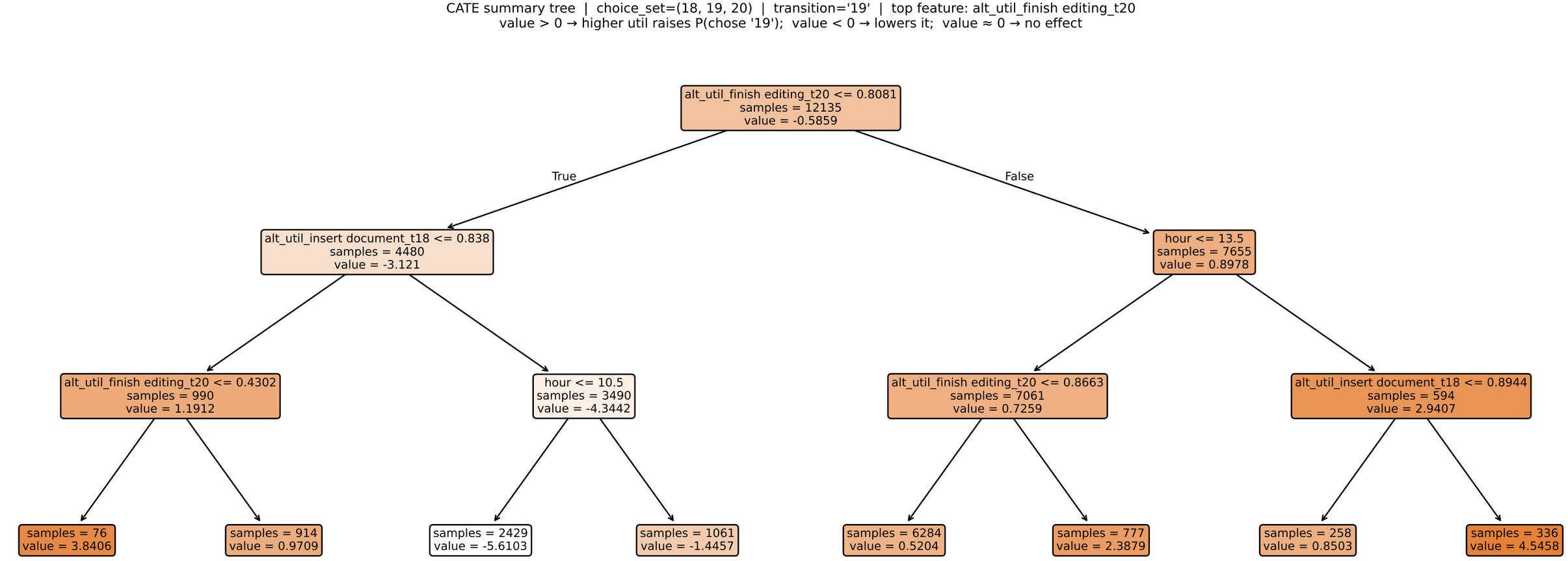


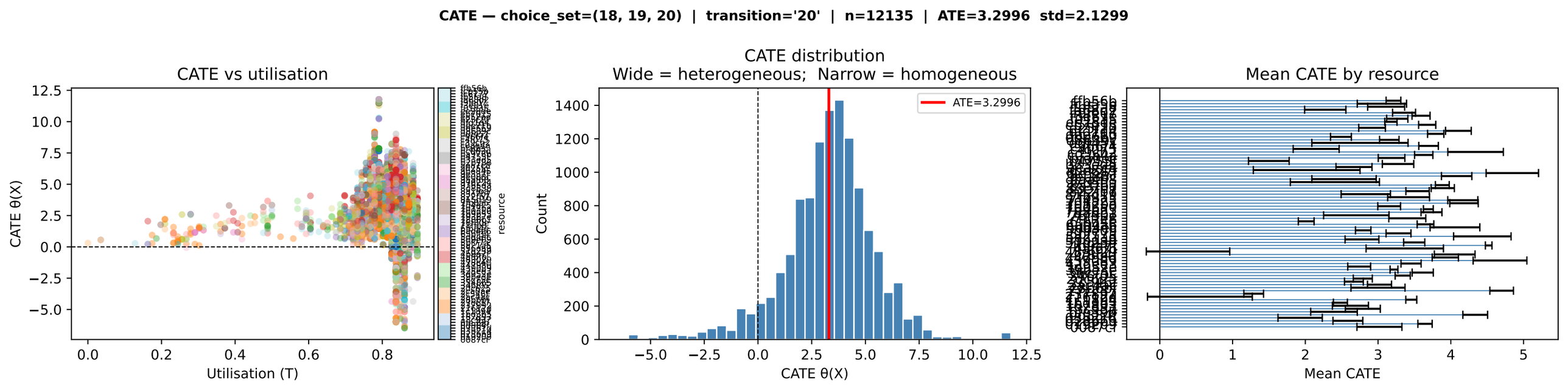


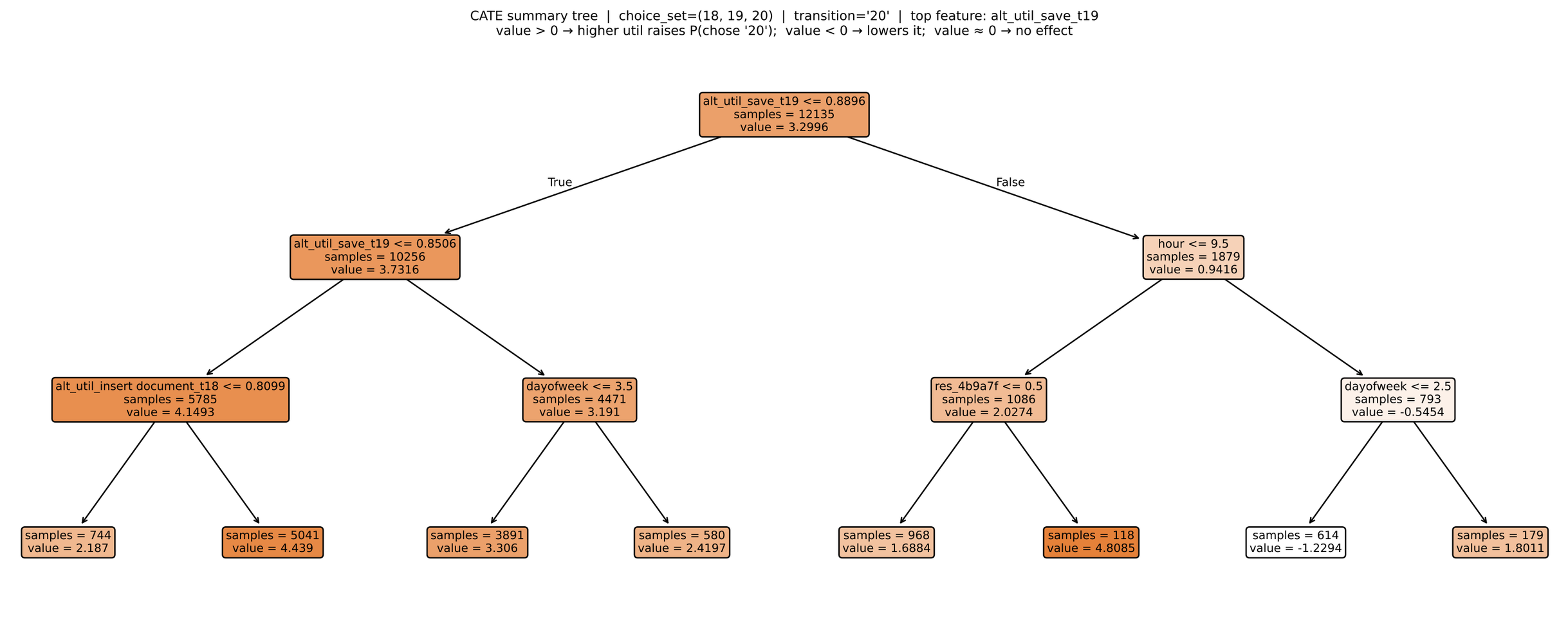



CATE summary tree | choice_set=(18, 19, 20) | transition='18' | top feature: alt_util_finish editing_t20
value > 0 → higher util raises P(chose '18'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t19 <= 0.8766
samples = 12135
value = 0.0376
True
False
alt_util_save_t19 <= 0.8506
samples = 8622
value = 0.4481
alt_util_finish editing_t20 <= 0.843
samples = 3513
value = -0.9697
hour <= 7.5
samples = 5785
value = 0.0634
alt_util_finish editing_t20 <= 0.8198
samples = 2837
value = 1.2325
hour <= 9.5
samples = 1999
value = -2.4819
hour <= 8.5
samples = 1514
value = 1.027
samples = 1313
value = -0.6983
samples = 4472
value = 0.2871
samples = 1478
value = 1.6648
samples = 1359
value = 0.7623
samples = 1085
value = -3.1985
samples = 914
value = -1.6313
samples = 518
value = -0.9028
samples = 996
value = 2.0306


CATE — choice_set=(18, 19, 20) | transition='19' | n=12135 | ATE=-0.5859 std=4.1088
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.5859
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(18, 19, 20) | transition='19' | top feature: alt_util_finish editing_t20
value > 0 → higher util raises P(chose '19'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t20 <= 0.8081
samples = 12135
value = -0.5859
True
False
alt_util_insert document_t18 <= 0.838
samples = 4480
value = -3.121
hour <= 13.5
samples = 7655
value = 0.8978
alt_util_finish editing_t20 <= 0.4302
samples = 990
value = 1.1912
hour <= 10.5
samples = 3490
value = -4.3442
alt_util_finish editing_t20 <= 0.8663
samples = 7061
value = 0.7259
alt_util_insert document_t18 <= 0.8944
samples = 594
value = 2.9407
samples = 76
value = 3.8406
samples = 914
value = 0.9709
samples = 2429
value = -5.6103
samples = 1061
value = -1.4457
samples = 6284
value = 0.5204
samples = 777
value = 2.3879
samples = 258
value = 0.8503
samples = 336
value = 4.5458


CATE — choice_set=(18, 19, 20) | transition='20' | n=12135 | ATE=3.2996 std=2.1299
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=3.2996
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(18, 19, 20) | transition='20' | top feature: alt_util_save_t19
value > 0 → higher util raises P(chose '20'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t19 <= 0.8896
samples = 12135
value = 3.2996
True
False
alt_util_save_t19 <= 0.8506
samples = 10256
value = 3.7316
hour <= 9.5
samples = 1879
value = 0.9416
alt_util_insert document_t18 <= 0.8099
samples = 5785
value = 4.1493
dayofweek <= 3.5
samples = 4471
value = 3.191
res_4b9a7f <= 0.5
samples = 1086
value = 2.0274
dayofweek <= 2.5
samples = 793
value = -0.5454
samples = 744
value = 2.187
samples = 5041
value = 4.439
samples = 3891
value = 3.306
samples = 580
value = 2.4197
samples = 968
value = 1.6884
samples = 118
value = 4.8085
samples = 614
value = -1.2294
samples = 179
value = 1.8011

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(0, 1)	0	14750	0.00497	0.073175	-0.244608	-0.038483	-0.000111	0.038963	0.617894	38.6	61.4	hour
(0, 1)	1	14750	0.126508	1.433083	-13.552544	-0.001466	0.004227	0.149228	1.923339	62.4	37.6	alt_util_begin editing_t0
(2, 3, 4, 5)	2	14749	0.985313	14.828215	-32.071366	-5.224298	-0.524841	1.873536	59.468374	46.6	53.4	alt_util_save_t3
(2, 3, 4, 5)	3	14749	-6.643038	20.99373	-81.144609	-3.850704	0.470721	1.543543	16.1254	54.1	45.9	alt_util_insert document_t2
(2, 3, 4, 5)	4	14749	-0.730771	2.338057	-17.172672	-0.497153	-0.194016	0.063099	5.727543	32.4	67.6	alt_util_insert document_t2
(2, 3, 4, 5)	5	14749	1.526212	5.635013	-16.278302	-0.136242	1.17888	2.907933	23.850974	73.6	26.4	hour
(6, 7, 8)	6	9601	0.006157	0.053677	-0.287847	-0.000372	0.000227	0.001063	0.574859	64.5	35.5	alt_util_save_t8
(6, 7, 8)	7	9601	0.001847	0.020731	-0.203781	-0.001236	0.000253	0.003513	0.105918	55.8	44.2	hour
(6, 7, 8)	8	9601	-0.007172	0.08002	-0.568912	-0.006459	-2.5e-05	0.003516	0.718512	48.8	51.2	hour
(9, 10, 11, 12, 13)	9	34043	0.007995	0.181549	-2.465944	0.001356	0.005964	0.017622	2.063513	82.1	17.9	alt_util_insert document_t10
(9, 10, 11, 12, 13)	10	34043	0.228443	1.39286	-10.008208	-0.034975	0.134122	0.420577	22.642443	70.9	29.1	alt_util_check_t13
(9, 10, 11, 12, 13)	11	34043	0.454035	7.286786	-28.803058	-3.687815	0.353426	4.379152	50.31211	52.2	47.8	alt_util_begin editing_t9
(9, 10, 11, 12, 13)	12	34043	4.304847	8.615309	-31.804304	-0.613132	3.809381	8.787627	51.889763	72.3	27.7	alt_util_save_t11
(9, 10, 11, 12, 13)	13	34043	-1.222846	6.508132	-39.358871	-4.073039	-0.45111	2.518022	25.008776	45.8	54.2	alt_util_insert document_t10
(14, 15, 16, 17, 21)	14	31436	2.566755	6.20859	-20.929431	0.30948	2.757537	5.815143	33.258612	76.8	23.2	alt_util_save_t15
(14, 15, 16, 17, 21)	15	31436	3.288921	6.413794	-18.619972	-0.68327	2.971998	6.483266	26.48726	69.8	30.2	hour
(14, 15, 16, 17, 21)	16	31436	-0.213058	5.467351	-41.423721	-2.614317	-0.304925	2.077683	34.799742	45.6	54.4	dayofweek
(14, 15, 16, 17, 21)	17	31436	0.170446	1.93222	-8.212501	-0.478244	0.107185	0.688554	7.610809	58.3	41.7	alt_util_insert document_t14
(18, 19, 20)	18	12135	0.037646	2.355544	-9.129498	-0.918831	0.088431	1.144407	11.09334	55.7	44.3	alt_util_finish editing_t20
(18, 19, 20)	19	12135	-0.585879	4.108758	-16.665567	-2.779135	0.489017	1.880372	13.249935	57.4	42.6	alt_util_finish editing_t20
(18, 19, 20)	20	12135	3.29958	2.129914	-6.482654	2.231418	3.486136	4.529798	11.770181	93.7	6.3	alt_util_save_t19

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md